

Machine Learning for Italian Non-Performing Loan Valuation: A Methodology Framework with Purged Cross-Validation

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Abstract

This paper is a methodology-protocol contribution, not an empirical study of Italian NPL portfolios. Version 2 (2026-08-18) differs from the deposited v1 in one structural respect: the synthetic generator’s recovery marginals are now *held out* against supervisory data rather than merely calibrated to aggregates, and the generator itself has been corrected in four places that v1 disclosed but did not repair. Section 6 is new and reports that test; Section 7 is a numbered erratum listing every changed number. We develop and demonstrate a machine-learning protocol for valuing Italian non-performing loans (NPLs) with three methodological commitments. First, we implement purged cross-validation with embargo periods [31], preventing the look-ahead bias that contaminates standard k -fold evaluation in financial time series. Second, we deploy LightGBM [28] with native categorical feature support, efficiently handling the high-cardinality judicial district variable (26 districts) that drives Italian NPL recovery heterogeneity. Third, we provide SHAP-based [33] global and local explanations; on the synthetic data the protocol surfaces `collateral_gbv_ratio`, `region_efficiency` and `vintage_years` as the top recovery-rate signals, which (as Section 3.3 documents) are recoveries of the injected DGP components rather than empirical discoveries—the same attribution machinery would become structurally informative for NPL servicers and banking regulators only when applied to real loan-level data. That ranking is itself unstable across generator revisions: v1 reported `is_secured`, `region_efficiency` and `guarantee_type`, which is a further reason to read SHAP on synthetic data as DGP wiring rather than as economics.

The protocol comprises five estimated models—an Automated Valuation Model (AVM) for property collateral, a recovery rate predictor, Weibull maximum likelihood estimation for recovery timing curves, and share-of-wallet and unsecured credit models—plus one deliberately simple illustrative pricing rule, a Nash midpoint Zone of Possible Agreement (ZOPA) for secondary market transactions, which we present as a placeholder rather than a serious pricing module (it omits servicing and funding costs, legal risk, and recovery-timing discounting). We demonstrate the protocol on a synthetic dataset of 100,000 loans (a 40-column loan table, of which 34 are modelling-eligible features) whose recovery marginals are calibrated to Banca d’Italia *Note di stabilità finanziaria e di vigilanza* N. 48 and whose closure timing and judicial gradient are calibrated to that note and to Ministero della Giustizia SIECIC court data; property values remain calibrated to the Osservatorio del Mercato Immobiliare (OMI).

We acknowledge upfront a feature-DGP-overlap circularity: the 24 features used in the recovery-rate model overlap 1:1 with the synthetic-DGP signal components, so the reported recovery R^2 measures the protocol’s ability to recover the calibrated DGP rather than generalization to real Italian NPL portfolios. That number is **DGP-internal** throughout this paper and is reported as $R^2 = 0.533$ out-of-time, *down* from v1’s 0.727; the decline is not a regression but the price of representing a closure channel (workout versus sale) that v1 could not represent at all and that a buyer does not observe at acquisition. Observing the channel is worth +0.1735 in R^2 —the value of that information, not a modelling gain. The AVM $R^2 = 0.499$ similarly reflects synthetic-DGP recovery, not benchmark performance against real OMI-anchored AVMs (which typically achieve $R^2 = 0.65$ – 0.75 on real Italian real-estate transactions). Executed Ridge and random-forest baselines still show no gradient-boosting edge worth claiming: on the recovery task LightGBM and Ridge are within 0.003 of one another (0.533 vs 0.530), and on the AVM Ridge remains ahead (0.528 vs 0.499). The share-of-wallet model warrants a separate and stronger caution: its target (income-to-GBV ratio) is computed directly from two candidate predictors, so a naive specification attains $R^2 \approx 0.98$ purely by target leakage; we exclude income and GBV from its feature set and report the honest out-of-time $R^2 = 0.175$ from exogenous covariates only.

Held-out anchoring (new in v2). The generator’s recovery marginals are calibrated only on Banca d’Italia exit years 2006–2019, so exit years 2020–2024 and entry cohorts after 2016 are genuinely out of sample *for the shipped generator*. On those held-out points the recovery-level layer attains a mean absolute error of 4.42 pp over 20 predictions (v1: 11.89 pp), which is the ceiling this source permits—the best available calibration rule attains 4.43 pp. The closure-timing layer, refuted out of sample in v1 by 42.86 pp, now misses by 6.92 pp; that is a sixfold improvement and it is *not* a vindication, because 25 of 27 points still under-predict, in the same direction as v1. The Weibull North–South closure ratio falls from 2.52 to 1.28 against the 1.325 measured across 132 tribunals in the 2025 SIECIC release.

What v2 does not establish. Nothing here makes the loan-level results empirical: Banca d’Italia publishes no loan-level data, so no test built on this source can validate a loan-level R^2 . The held-out result validates the generator’s recovery *marginals*; its structure is untested and its timing layer, while much improved, remains biased in one direction. Real loan-level validation is still pre-registered and still requires an access route we do not yet hold.

Keywords: Non-performing loans, Machine learning, LightGBM, SHAP, Purged cross-validation, Italian banking, Recovery rate, Weibull, Automated valuation model

1. Introduction

The Italian banking system experienced the largest non-performing loan (NPL) crisis in European history. NPL stocks peaked at EUR 341 billion in 2015, representing approximately 17% of total loans—more than four times the EU average [19]. The subsequent resolution, driven by the GACS securitization guarantee scheme [39], ECB supervisory pressure [8], and an active secondary market, reduced NPL stocks to approximately EUR 50 billion by 2024. This unprecedented workout created a €40+ billion secondary market for distressed debt, requiring sophisticated valuation tools.

Accurate NPL valuation is critical for three constituencies. First, *banks* must estimate expected loss under Basel III/IV internal ratings-based (IRB) frameworks [40, 41]. Second, *NPL servicers and investors* in the secondary market need granular loan-level recovery predictions to price portfolios. Third, *regulators* require transparent, interpretable models to assess provisioning adequacy [5, 35].

Machine learning (ML) methods have demonstrated significant improvements over traditional credit risk models [1, 36] in predicting default [29] and loss given default [32, 24]. However, the application of ML to NPL valuation faces three specific challenges that this paper addresses:

1. **Cross-validation leakage.** Standard k -fold cross-validation assumes independent and identically distributed (i.i.d.) observations, violating the temporal dependence structure of financial data. Lopez de Prado [31] demonstrates that this leads to inflated performance metrics and overfit models. We implement *purged cross-validation* with embargo periods, demonstrated here on a synthetic, aggregate-calibrated Italian-NPL dataset (we make no priority claim over real Italian NPL data, which this paper does not use).
2. **High-cardinality categorical features.** Italian NPL recovery is heavily driven by judicial district. v1 motivated this point with median foreclosure durations ranging from roughly 24 months in Milan to 84+ months in Calabria, attributed to published Banca d’Italia and Ministero della Giustizia statistics on civil-execution proceeding durations. *Those figures do not reproduce and we do not assert them here*: no such per-district durations appear in any current public release, and the citation v1 gave for them does not resolve (erratum item E9). v2 therefore rests the judicial dimension on the *relative* SIECIC disposition-time gradient instead—a South/North median ratio of 1.325 across 132 tribunals in the 2025 release—used as a cross-district ordering and never as an absolute duration. One-hot encoding these 26 districts is inefficient and loses ordinal structure. We use LightGBM’s native categorical feature handling [28], which partitions categories optimally at each tree split.
3. **Regulatory interpretability.** Banking regulators increasingly require model explainability [9]. We provide SHAP (SHapley Additive exPlanations) decompositions [33, 34] at both global and local levels, directly mapping model predictions to the economic drivers that practitioners can act upon.

Contributions. This paper makes four *methodological* contributions, all demonstrated on synthetic data. First, we assemble an integrated valuation protocol—five models (AVM, recovery rate, Weibull recovery timing, share-of-wallet, unsecured recovery) plus one illustrative pricing rule (ZOPA)—covering property valuation, recovery rates, recovery timing, payment capacity, and a stylized secondary-market price. Second, we hold the generator’s recovery marginals out against supervisory data (Section 6): calibrating only on Banca d’Italia exit years 2006–2019 leaves 20 recovery-level and 27 closure-timing points that the shipped generator has never seen, converting the paper’s central caveat from a disclaimer into a measurement. Third, we provide SHAP-based feature attribution and document, via an injected-vs-learned audit (Section 3.3) and a κ sensitivity sweep (Section 5.5), exactly

which “drivers” and which accuracy figures are recoveries of calibration inputs rather than discoveries. Fourth, we release a complete replication package, integrity-sealed so that every file it ships is covered by a SHA-256 manifest that verifies (Appendix [Appendix A](#)), together with a numbered erratum (Section [7](#)) recording every v1 figure this revision changes and the two it discloses but does not fix.

What changed in v2, in one paragraph.. v1 was calibrated to Italian aggregates and said so; it had no external test, and four defects it disclosed in late revisions went unrepaired. v2 adds the missing closure-channel dimension, re-anchors a closure-timing layer that was refuted out of sample by 42.86 pp, gives `ltv` the causal path to recovery it never had, and turns the noise constant κ into a reported sensitivity. The held-out recovery-level error falls from 11.89 to 4.42 pp and the timing error from 42.86 to 6.92 pp. Three things move the other way and are reported as such: the DGP-internal recovery R^2 falls from 0.727 to 0.533 because the generator now represents something a buyer cannot observe; the serial-dependence ablation no longer supports the claim that purging binds; and the timing layer, though six times closer, still under-predicts at 25 of 27 held-out points.

This paper is part of a broader research program on private credit systemic risk. The parent paper [\[48\]](#) develops the K-A-T framework linking funding constraints, network amplification, and policy backstop credibility. A companion paper [\[46\]](#) maps GP-LP-Bank interconnections using DebtRank. A third companion [\[47\]](#) analyzes the retailization of private credit. The present paper provides the credit valuation methodology dimension.

The remainder of this paper is organized as follows. Section [2](#) reviews related work. Section [3](#) describes the data sources and synthetic generation methodology. Section [4](#) presents the modeling framework. Section [5](#) reports results. Section [8](#) discusses implications, and Section [9](#) concludes.

2. Literature Review

2.1. Machine Learning in Credit Risk

The application of ML to credit risk has evolved rapidly from simple logistic regression [\[1\]](#) through ensemble methods [\[11, 21\]](#) to gradient boosting machines [\[13, 28\]](#). Gu et al. [\[23\]](#) provide a comprehensive comparison of ML methods in asset pricing, finding that gradient-boosted trees and neural networks significantly outperform linear models.

In the specific context of credit default and LGD prediction, Khandani et al. [\[29\]](#) demonstrate that classification trees and logistic regression ensembles outperform traditional credit scoring. Loterman et al. [\[32\]](#) benchmark 24 algorithms for LGD modeling, finding that support vector machines and neural networks achieve the best performance. Barbaglia et al. [\[10\]](#) apply ML to European loan default forecasting, highlighting the importance of macroeconomic features.

2.2. NPL Valuation in Italy

The Italian NPL market has generated a substantial empirical literature. Published Banca d’Italia statistics document mean recovery rates of roughly 45% for secured and materially lower rates for unsecured exposures (the Bank of Italy’s annual “Bad loan recovery rates” notes report secured recovery near 40–50%), alongside a factor-of-three North–South

difference in foreclosure-proceeding duration widely attributed to civil-justice efficiency; the public-aggregate sources used to calibrate the synthetic dataset are listed in Section 3.

The GACS scheme [39] catalyzed the development of a secondary NPL market, generating demand for granular valuation tools [42, 12].

2.3. Interpretable ML and SHAP

Lundberg and Lee [33] unify several interpretability methods under the SHAP framework, building on Shapley values [45]. Lundberg et al. [34] extend this to tree-based models with exact TreeExplainer computation. Molnar [37] provides a comprehensive guide to interpretable ML methods.

In banking regulation, the ECB [8, 9] and EBA [5] increasingly require model interpretability for internal models used in provisioning and capital calculation. SHAP decompositions directly address this regulatory requirement.

2.4. Cross-Validation in Financial ML

Lopez de Prado [31] identifies cross-validation leakage as the primary source of false discoveries in financial ML. The key insight is that financial time series exhibit serial correlation, causing standard k -fold to include future information in training folds. Purged cross-validation removes observations temporally adjacent to test folds, while the embargo period accounts for serial correlation decay. Bailey and Lopez de Prado [7] extend this analysis to Sharpe ratio estimation and backtest overfitting.

3. Data

Synthetic data disclosure. This paper uses a synthetic loan-level dataset calibrated to Italian aggregate statistics. No proprietary or real borrower-level data is used. We adopt this approach because (a) Italian NPL loan-level data is proprietary to banks and servicers, (b) our contribution is methodological—demonstrating ML best practices for NPL valuation—rather than empirical, and (c) the synthetic generation process is fully documented and reproducible. This follows recommendations for methodology papers in credit risk [27, 4].

3.1. Aggregate Data Sources

We calibrate the synthetic dataset using three Italian public data sources:

Banca d'Italia.. NPL aggregate statistics from the Financial Stability Report [19, 18, 17]: total NPL stock (sofferenze) and unlikely-to-pay (UTP/inadempienze probabili) by year (2008–2025), coverage ratios for secured and unsecured exposures, and regional and sectoral distribution.

Osservatorio del Mercato Immobiliare (OMI).. Property price indices from the Agenzia delle Entrate [16]: residential, commercial, and land prices per square meter by Italian macrozone (Nord-Ovest, Nord-Est, Centro, Sud, Isole) and semester (2008–2025). Cross-validated against the ISTAT house price index (IPAB) [26].

Ministero della Giustizia.. v1 cited [15] for “average and median duration for real estate foreclosures across 26 major tribunal districts”. **No such figures exist in the current public monitoring release** (erratum item E9). What the Ministry publishes, per appeal-court district, is the SIECIC monitoring workbook: case flows (*sopravvenuti / definiti*) for ESECUZIONI IMMOBILIARI and an age-of-pending stratigraphy—clearance and stock, not realised duration. v2 therefore uses the SIECIC Q4-2025 release (25 of 26 appeal-court districts obtained; Catanzaro returned HTTP 404) and derives a CEPEJ disposition time, $12 \times$ pending/resolved, for 132 tribunals. That proxy is used only as a *relative* cross-district gradient, never as an absolute duration; the level comes from Banca d’Italia (below).

Banca d’Italia, Note di stabilità finanziaria e di vigilanza *N. 48 (new in v2)*.. “I tassi di recupero delle sofferenze nel 2024” (18 December 2025), statistical annex, SHA-256 73651aa96c85f30f.... Underlying data: the Centrale dei Rischi. This is the source that makes v2’s held-out test possible. Three of its tables are used: Tav. A2 (recovery rate by collateral $\times$ closure channel, exit years 2006–2024), Tav. A1 (the incidence of each of those four cells), and Tav. 4 (cumulative share of positions closed within 1...6 years, by entry cohort 2006–2023). The annex carries *no* territorial dimension of any kind, which is why the judicial-heterogeneity story remains untested on the recovery side. A parser with 70 assertions against the note’s own published figures and cross-table identities is part of the replication package.

3.2. Synthetic Loan Generation

We generate 100,000 synthetic NPL loans as a 40-column loan table (of which 34 are modelling-eligible features; the recovery-rate model uses 24 and the AVM 12, with the remainder identifiers, dates, targets, and the train/test split flag), using a structural data-generating process (DGP) that embeds realistic cross-feature correlations. Feature distributions are calibrated both to Italian aggregate statistics and to operational NPL platform parameters (forced sale discounts, regional efficiency scores, procedure type distributions).

The calibration window is deliberately truncated (new in v2).. Every Banca d’Italia constant in the generator is computed on exit years 2006–2019 and entry cohorts up to 2016 only. Exit years 2020–2024 and the later cohorts are withheld. This costs nothing in fidelity—the marginals are flat enough that the extra years barely move the targets—and it buys the one thing the v1 design could not have: a test set the shipped generator has never seen. Section 6 scores it there. Calibrating on the full series, as would be natural, would have converted that test into a round-trip.

Weighting basis.. The annex’s published *Media* rows are amount-weighted pooled rates; pooling across 2006–2024 bakes in the composition of the 2017–2021 sale wave. v2 calibrates instead to *equal-year* means of the annual rates over the calibration window, which is also exactly the object the held-out benchmark in Section 6 uses, so generator and benchmark are the same construct. Both weightings are computed and shipped; Table 2 reports how much the choice matters. A residual mismatch remains and is not removed by this choice: the synthetic book is a book of *positions*, while every incidence and rate table in the annex is amount-weighted. Over the calibration window the sold share is 0.294 by amount against 0.318 by number.

The calibration targets are:

- **Regional distribution:** 60% South/Islands, 25% Center, 15% North — a deliberate over-weighting of high-default-rate Southern districts that reflects the documented North–South NPL-*ratio* gradient (bad-loans-to-total-loans). Note this is the *ratio* gradient, not the NPL *stock* share: by volume the NPL stock is in fact concentrated in the North, where total lending is larger (Banca IFIS Market Watch). The synthetic over-sampling of Southern districts is a modelling choice to stress the judicial-heterogeneity signal, not a claim about the national stock split.
- **Sector distribution:** 40% corporate, 35% SME, 25% retail (matching Bdl sector breakdown)
- **Property values:** Log-normal by macro-zone, calibrated to OMI price data, adjusted for property condition, energy class, and vintage year
- **Closure channel (new in v2):** each position closes either *in via ordinaria* (worked out) or by sale to a third party. v1 had no channel dimension at all, and the sale channel is far too large to omit—though its size is not a single number, and we decline to quote one. Over the calibration window the sold share is 0.294 of closures by amount and 0.318 by number under equal-year weighting, against 57.1% and 42.0% under the amount-weighted pooling this paper elsewhere declines to use (Table 2); year by year it climbs from 2.6% of closed amounts in 2006 to 86.2% in 2018. Under every weighting it is a first-order share of the book, which is what makes its absence from v1 a modelling defect rather than a rounding one. The conditional incidences are Tav. A1 equal-year means over the calibration window: $P(\text{sold} \mid \text{secured}) = 0.2761$, $P(\text{sold} \mid \text{unsecured}) = 0.3024$. The realised sold share of the generated book is 28.0%.
- **Recovery rates:** Beta-distributed. The *level* of each of the four collateral $\times$ channel cells is its Tav. A2 equal-year mean over the calibration window—56.29%, 38.37%, 38.27% and 18.87% respectively. v1 instead targeted two round numbers, 0.45 secured and 0.15 unsecured, that appear in no Banca d’Italia table; the second of these is the single largest error v2 corrects (erratum item E7). The *cross-section* within each cell is an 11-component additive structural signal: judicial efficiency, collateralisation, property condition, income capacity, creditor count, procedure type, guarantee type, vintage decay, coverage ratio, forced-sale discount, and—new in v2—LTV, which in v1 entered no component at all (erratum item E1). Each cell is then recentred on its Banca d’Italia target, so the marginals sit on the published rate by construction rather than approximately. That recentring *replaces* v1’s twelfth component, an additive base of 0.50 secured against 0.18 unsecured: the collateral level now lives in the cell offset rather than in a within-cell coefficient, which is why `is_secured` falls down the attribution ranking in Section 5.6 without anything economic having changed.
- **Recovery timing:** Weibull-distributed, anchored to the Tav. 4 closure curve of entry cohort 2016 (shape 1.0219, scale 1.9708 years, implied median 16.5 months). v1 used shape 1.5 around a national median of 54 months, implying that only 7.0% of positions close within a year against a realised 42.0–64.8% (erratum item E8). The cross-district

gradient is the SIECIC 2025 disposition-time multiplier, normalised to book mean 1 so it cannot move the national level, and applied to worked-out positions only—a sold position closes on its transfer date, which no court controls.

- **Additional features:** Property surface area (sqm), energy class, condition score, guarantee type, procedure type, forced sale discount, regional efficiency encoding, NBV/GBV ratio, LTV, realisable-collateral share, accrued-interest ratio, and derived log-transforms and ratios
- **Temporal split:** Loans defaulted pre-2022 (75,525 loans) for training; 2022–2025 (24,475) for out-of-time testing

Table 1 validates the calibration by comparing synthetic dataset moments to aggregate targets. **Matching a target there is a calibration check, not evidence;** the evidence is Section 6, which scores the same generator on years it never saw.

Table 1: Calibration Validation: Synthetic Dataset vs Italian Aggregate Statistics

Moment	Synthetic	Target	% Diff
Pct South+Islands	0.5991	0.5970	0.3600
Pct Center	0.2503	0.2515	-0.4700
Pct North	0.1505	0.1515	-0.6300
Pct corporate	0.4020	0.4030	-0.2400
Pct SME	0.3507	0.3500	0.2000
Pct retail	0.2473	0.2470	0.1100
Recovery rate (secured, worked out)	0.5633	0.5629	0.0600
Recovery rate (secured, sold)	0.3847	0.3837	0.2500
Recovery rate (unsecured, worked out)	0.3820	0.3827	-0.1800
Recovery rate (unsecured, sold)	0.1900	0.1887	0.6900
Sold share (secured)	0.2740	0.2761	-0.7300
Sold share (unsecured)	0.2955	0.3024	-2.3000
Mean property value (Nord-Ovest)	139285.6699	145000.0000	-3.9400
Mean property value (Sud)	58699.6830	62000.0000	-5.3200
Closed within 1 year	0.4072	0.3934	3.5000
Median closure time (months)	16.0000	16.5200	-3.1500
Closure time South/North ratio	1.2846	1.3249	-3.0400

Note: Synthetic dataset of 100,000 loans calibrated to Banca d’Italia, OMI, and Ministero della Giustizia aggregate statistics. Recovery-rate and closure-channel targets are Banca d’Italia Note 48, Tav. A1 and Tav. A2, equal-year means over exit years 2006–2019; closure-timing targets are Tav. 4, entry cohort 2016; the district gradient is the 2025 SIECIC disposition-time ratio. Exit years 2020–2024 are deliberately withheld from the calibration so that the held-out test remains out of sample for this generator. Matching a target here is a calibration check, not evidence: the out-of-sample evidence is the 2020–2024 test reported separately.

Channel or collateral: which margin is larger?. It is tempting to report that the closure channel dominates collateral as a driver of recovery, because under the annex’s own amount-weighted pooling it does. Table 2 shows why that claim should not be made. Under equal-year weighting on the calibration window the two families of margin are within 1.58 pp of each other; the apparent channel dominance in the pooled column is composition, since sold amounts concentrate in the low-recovery 2017–2021 sale wave. **The defensible statement is that the two margins are comparable in size and that v1 modelled only one of them.**

Table 2: Recovery-rate margins under three weightings of the same Banca d’Italia Tav. A2 cells (percentage points)

Margin	Equal-year 2006–2019	Equal-year 2006–2024	Pooled, amount-weighted 2006–2024
Channel, within secured	17.92	16.43	19.16
Channel, within unsecured	19.39	18.29	18.96
Collateral, within worked out	18.03	15.90	15.84
Collateral, within sold	19.50	17.75	15.64

Note: All four margins are differences of the same four Tav. A2 cells; only the weighting across exit years differs. Column 1 is the window the generator is calibrated on. Only under amount-weighted pooling does the channel margin exceed both collateral margins, and that ordering is driven by the concentration of sold *amounts* in the 2017–2021 sale wave rather than by any within-year fact. On the calibration window the largest separation between any channel margin and any collateral margin is 1.58 pp.

3.3. DGP faithfulness: what is injected vs. what is learned

Because the recovery-rate target is constructed from an additive 11-component structural signal on top of a per-cell offset, the model’s task on this synthetic dataset is to *recover* that injected signal rather than to discover unknown structure. Table 3 makes this explicit: it maps each injected DGP component to the recovery-model feature that carries it and to that feature’s SHAP rank in the results below (Table 9). The 1:1 overlap means the headline recovery R^2 and the SHAP “finding” that collateral_gbv_ratio, region_efficiency and vintage_years dominate are **DGP-recovery sanity checks**—they confirm the protocol correctly identifies the signal it was given against the irreducible Beta-noise floor, but they are *not* empirical discoveries about real Italian NPL portfolios. The v1 book’s top three were a different set entirely (is_secured, region_efficiency, guarantee_type), which is the same point made twice. We therefore frame the entire results section as a DGP-faithfulness demonstration: the contribution is the methodology protocol and its provable recovery of a known signal, not the recovered numbers themselves.

The reported R^2 is set by one generator constant.. The preceding paragraph says the headline recovery R^2 measures DGP recovery rather than generalisation. We can be more specific about what fixes its *level*, because the level is not an emergent property of the protocol at all. The recovery target is drawn as a Beta variate whose concentration κ sets $\text{Var}(Y | X) = t(1 - t)/(\kappa + 1)$ and therefore the attainable ceiling. v1 hard-coded $\kappa = 20$ with the

accompanying comment recording that it had been raised from 12 “→ stronger signal-to-noise”.

What can and cannot be said about $\kappa = 20$. It is not supported by any public aggregate, and it is not refuted by one either. The Banca d’Italia annex publishes amount-weighted aggregates only; it carries no loan-level dispersion information whatsoever, so κ —a loan-level dispersion parameter—can be neither calibrated nor falsified from it. Two readings of $\kappa = 20$ fail in opposite directions. Read as a loan-level band it covers every held-out point with an interval roughly 36 pp wide, which cannot reject anything. Read as a portfolio-mean band it rejects overwhelmingly against a 100,000-loan book. The nearest testable analogue—the concentration whose implied dispersion matches the year-to-year variation of the aggregate around its own training mean—is 46 to 128, but that is a different object and should not be presented as “the correct κ ”.

v2’s response is not to invent a value. κ is now an argument rather than a constant, the default is left at 20 for comparability with the deposited number, and Section 5.5 reports which outputs move with it. The short answer is that the DGP-internal R^2 moves across most of its usable range while the held-out result of Section 6 does not move at all. We also flag the asymmetry in the generator’s own design: the coverage-ratio Beta *is* calibrated to Banca d’Italia aggregates, so the machinery for anchoring a Beta parameter to real data was present and was not applied to the one parameter that sets the reported R^2 .

4. Methodology

4.1. Purged Cross-Validation

Standard k -fold cross-validation randomly partitions observations into k folds, implicitly assuming i.i.d. data. In financial time series with serial correlation, this allows information from future periods to leak into training folds, inflating out-of-sample performance estimates [31, 3].

Purged cross-validation addresses this by:

1. **Temporal sorting:** Observations are sorted by default date before partitioning.
2. **Purging:** For each test fold, training observations within a window of $\epsilon \cdot N$ observations of the test fold boundaries are removed, where ϵ is the embargo percentage (we use $\epsilon = 0.01$, i.e., 1% of the dataset).
3. **Embargo:** An additional gap of $\epsilon \cdot N$ observations is imposed after the test fold end to account for serial correlation decay.

For our dataset with $N = 75,525$ training observations, the embargo removes approximately 755 observations per boundary, totaling $\sim 1,510$ observations per fold (2% of training data). This modest cost is justified by the elimination of temporal information leakage. *Caveat on the baseline synthetic demonstration.* Because the baseline synthetic recovery target is drawn i.i.d. conditional on its features, it contains no serial dependence for the purge to remove; on the baseline dataset purged and standard k -fold cross-validation are statistically indistinguishable, and the purge is non-binding by construction. Rather than

Table 3: DGP faithfulness: what is injected vs. what is learned (recovery-rate model, v2.1 book)

#	Injected DGP signal component	Recovery-model feature	SHAP rank
2	Judicial efficiency $(\text{eff} - 1.5) \times 0.06$	<code>region_efficiency</code>	2
3	Collateralisation $(\min(\text{coll}/\text{GBV}, 2) - 1.0) \times 0.15$; in v2.1 the ratio is <code>pledged_share/ltv</code>	<code>collateral_gbv_ratio</code>	1
4	Property condition $(\text{score} - 2) \times 0.015$	<code>condition_score</code>	8
5	Income/GBV capacity $\text{clip}(\log(1+\cdot) \times 0.03, \pm 0.04)$	<code>income_gbv_ratio</code>	12
6	Creditor count $-\min(n-1, 6) \times 0.008$	<code>n_creditors</code>	9
7	Procedure type $(-0.01 \text{ to } +0.04)$	<code>procedure_type</code>	4
8	Guarantee type $(-0.02 \text{ to } +0.04)$	<code>guarantee_type</code>	5
9	Vintage decay $-\text{years} \times 0.005$	<code>vintage_years</code> , <code>years_since_default</code>	3, 13
10	Coverage ratio $(\text{cov} - 0.55) \times 0.08$	<code>coverage_ratio</code>	10
11	Forced-sale discount $-(\text{fsd} - 0.20) \times 0.12$	<code>forced_sale_discount</code>	17
12	LTV $-(\text{ltv} - 0.68) \times \beta_{\text{ltv}}$ (new in v2.1)	<code>ltv</code>	6
—	Per-cell recentring on the Banca d’Italia Tav. A2 collateral $\times$ channel target (replaces v1’s additive secured/unsecured base)	<code>is_secured</code>	14
—	Irreducible Beta noise (concentration $\kappa = 20$, an argument rather than a constant)	— (sets the R^2 ceiling, 0.7333 closed-form)	—

Note: Each row maps one additive component of the synthetic recovery-rate data-generating process (`synthetic_generator.py`) to the recovery-rate model feature that carries it and to that feature’s mean-|SHAP| rank in Table 9. Because the 11 injected components overlap 1:1 with model features, the SHAP rankings are recoveries of injected inputs, not discovered facts about real Italian NPLs. The reported recovery $R^2 = 0.533$ therefore measures DGP recovery against the irreducible Beta-noise floor, not generalization.

Two v2.1 changes are visible here and are the reason the ranking moved. First, v1’s Component 1 — an additive base of 0.50 secured against 0.18 unsecured — no longer exists: each collateral $\times$ channel cell is recentred on its published Tav. A2 target instead, so the secured/unsecured level sits in the cell offset rather than in a within-cell coefficient, and `is_secured` falls out of the top of the ranking accordingly. Second, `collateral_gbv_ratio` is now genuinely collateral over exposure; in v1 it reduced algebraically to $1/(1 + \text{accrued})$ and measured accrued interest, not collateralisation. Neither movement is economics.

leave the leakage-correction claim merely asserted, we test it directly with a controlled serial-dependence ablation (Section 5.4): we inject a vintage-clustered autocorrelated latent state—present in real Italian NPL panels but omitted by the baseline DGP—and measure the gap between standard and purged k -fold as a function of the autocorrelation. On the v2 book that ablation establishes only the first half of what v1 read into it: the purge is non-degrading, and the leakage gaps at $\rho > 0$ are at the noise floor and do not support the claim that purging binds (Section 5.4, erratum item E12). The Track A/B real-data evaluation pre-registered in Section 8.4 (timeline in Section 8, 2027-Q1) is where a binding case, with overlapping-horizon labels, could be tested.

4.2. *LightGBM with Native Categoricals*

We use LightGBM [28] as the primary model for three reasons: (a) competitive predictive performance on tabular data [23], (b) native categorical feature support avoiding one-hot encoding overhead, and (c) efficient SHAP TreeExplainer computation [34].

LightGBM’s categorical feature handling works by finding the optimal partition of categories at each tree split using gradient statistics, rather than treating categories as unordered binary variables. This is particularly important for judicial district (26 levels) and region (20 levels), where one-hot encoding would create sparse, high-dimensional features.

Hyperparameters are tuned per model: learning rate 0.05, 63 leaves, 400–500 estimators, 80–85% subsample, with L1/L2 regularization 0.03–0.05. The AVM and recovery rate models use deeper trees (max depth 7) to capture complex feature interactions. These are standard choices for mid-to-large tabular datasets [28, 25].

4.3. *Model Specifications*

The framework comprises five estimated models and one illustrative pricing rule:

Model 1: Automated Valuation Model (AVM). Predicts property collateral value $\hat{V}_i$ as a function of 12 features: region, macro-zone, property type, judicial district, origination year, secured status, property surface area, condition score, energy class score, regional efficiency, and debtor type. The AVM serves as an input to both the recovery rate model and the ZOPA calculator.

Model 2: Recovery Rate. Predicts the fraction of gross book value recovered, $\hat{r}_i \in [0, 1]$, using 24 features including secured status, collateral and book values, judicial district, debtor characteristics, coverage ratio, guarantee type, procedure type, regional efficiency encoding, condition and energy scores, forced sale discount, NBV/GBV ratio, LTV, and derived log-transforms and ratios. This is the core model of the framework.

Model 3: Weibull Recovery Curves. Estimates the time-to-recovery distribution by fitting Weibull distributions via maximum likelihood [30, 14]:

$$f(t; k, \lambda) = \frac{k}{\lambda} \left(\frac{t}{\lambda} \right)^{k-1} e^{-(t/\lambda)^k}$$

where k is the shape parameter and λ the scale parameter. We fit separate curves for secured/unsecured, North/Center/South, and corporate/SME/retail segments.

Model 4: Share-of-Wallet.. Predicts the debtor’s payment capacity ratio (income relative to outstanding exposure), used to assess voluntary repayment potential.

Model 5: Unsecured Credit Recovery.. A specialized recovery model trained only on unsecured loans, where the absence of collateral makes debtor characteristics the dominant predictors.

Illustrative pricing rule: ZOPA.. As a deliberately simple placeholder—*not* a co-equal model—we compute the symmetric Nash-midpoint price [38, 44] for secondary-market transactions:

$$p^* = \frac{p_{\text{buyer}} + p_{\text{seller}}}{2} \quad \text{if } p_{\text{buyer}} \geq p_{\text{seller}}$$

where p_{buyer} is the buyer’s reservation price (from the recovery model) and p_{seller} is the seller’s reservation price (from AVM discounted by coverage ratio). The Zone of Possible Agreement exists when the buyer’s valuation exceeds the seller’s. This is an identity-level rule: it omits servicing cost, funding cost, legal risk, recovery-timing discounting, and any IRR/return-target structure, and equal bargaining power is assumed rather than estimated. A serious secondary-market pricing module incorporating these frictions is out of scope for v1; we include ZOPA only to show where the recovery and AVM outputs would feed such a module.

4.4. SHAP Explanations

We compute SHAP values using TreeExplainer [34], which provides exact Shapley value computation in polynomial time for tree-based models. For each prediction $\hat{y}_i$:

$$\hat{y}_i = \phi_0 + \sum_{j=1}^p \phi_{ij}$$

where ϕ_0 is the base value (mean prediction) and ϕ_{ij} is the SHAP contribution of feature j for observation i . We report both global importance (mean $|\phi_j|$ across observations) and local explanations for individual loans.

4.5. Out-of-Time Validation

The primary evaluation metric is out-of-time (OOT) validation: models are trained on loans defaulted before 2022 and evaluated on loans defaulted 2022–2025. This temporal separation prevents any form of future information leakage and is the most stringent test of model generalizability in a financial context [31].

5. Results

Every R^2 in this section is DGP-internal.. It measures how well LightGBM recovers a data-generating process written down in this repository, on a book the same repository generated. It is not an out-of-sample result and must not be read as one. The paper’s out-of-sample results are in Section 6, they concern the generator rather than the models, and they are reported in percentage points of recovery rate rather than in R^2 .

5.1. Model Performance

Table 4 reports cross-validated and out-of-time performance for all four LightGBM models on the v2 book. The recovery-rate model attains purged CV $R^2 = 0.529$ and the AVM $R^2 = 0.536$, the latter reflecting the enriched feature set including property condition, energy class, and surface area. The share-of-wallet model, whose target (income-to-GBV ratio) is by construction a deterministic function of two candidate predictors (debtor income and gross book value), attains purged CV $R^2 = 0.190$ once those two definitionally-related features are excluded; a naive specification that retains them attains $R^2 \approx 0.98$ purely by target leakage, which we therefore avoid.

The recovery R^2 falls from 0.727 to 0.533, and that is the intended outcome.. Three of the four models score lower on the v2 book than on v1’s: recovery $0.727 \rightarrow 0.533$, share-of-wallet $0.224 \rightarrow 0.175$, unsecured $0.339 \rightarrow 0.219$ (the AVM is unchanged at 0.499, since nothing in the property-value block was touched). The cause is the closure channel added in Section 3. A position’s recovery now depends on whether it is worked out or sold, and **that outcome is not a feature of the model**, because a buyer pricing a portfolio at acquisition does not know which positions the servicer will eventually sell. Adding `closure_channel` to the feature set raises the recovery R^2 by 0.1735 (to 0.7062). We report that gap as the *value of information* about the exit route, not as a modelling improvement, and we keep the lower number as the headline because it is the one that corresponds to the decision the protocol is for.

The general point is worth stating once: a synthetic-data R^2 moves whenever the generator becomes more faithful, and it usually moves down. A rising R^2 across generator revisions would be evidence of a simpler simulator, not of a better model.

Table 4: Out-of-Sample Model Performance

Model	CV R^2	CV MAE	OOT R^2	OOT MAE	OOT RMSE	N train	N test
AVM	0.536	27196.825	0.499	29457.764	40345.364	75525.000	24475.000
Recovery Rate	0.529	0.109	0.533	0.110	0.134	75525.000	24475.000
Share-of-Wallet	0.190	15.087	0.175	16.185	42.627	75525.000	24475.000
Unsecured Credit	0.243	0.116	0.219	0.119	0.146	22145.000	7298.000

Note: Purged cross-validation (5-fold, 1% embargo) on pre-2022 training set. Out-of-time (OOT) test on 2022–2025 loans. LightGBM with native categorical features and proprietary-model-calibrated hyperparameters. 100,000 synthetic loans (40-column loan table; 34 modelling-eligible features).

5.2. Out-of-Time Validation

Table 5 provides detailed OOT results. The recovery-rate model’s OOT R^2 of 0.533 is marginally *above* its CV estimate of 0.529, so there is no evidence of overfitting; on the v1 book the same comparison ran the other way (0.746 CV against 0.727 OOT). The AVM shows a modest CV-to-OOT gap (0.536 to 0.499). The share-of-wallet model, trained on exogenous covariates only, attains OOT $R^2 = 0.175$ —an honest measure of how much payment capacity is predictable from non-definitional features. The unsecured credit model achieves OOT $R^2 = 0.219$, reflecting the intrinsic difficulty of recovery prediction without a collateral signal even with the enriched feature set.

Table 5: Out-of-Time Validation: Train Pre-2022, Test 2022–2025

Model	Train Period	Test Period	N train	N test	R^2	MAE	MAPE (%)
AVM	Pre-2022	2022-2025	75525.000	24475.000	0.499	29457.764	42.160
Recovery Rate	Pre-2022	2022-2025	75525.000	24475.000	0.533	0.110	41.929
Share-of-Wallet	Pre-2022	2022-2025	75525.000	24475.000	0.175	16.185	209.149
Unsecured Credit	Pre-2022	2022-2025	22145.000	7298.000	0.219	0.119	68.020

Note: Primary evaluation metric. Purged CV scores reported separately.

5.3. Baseline comparison

Table 6 reports the executed comparison of LightGBM against a Ridge linear baseline and a random-forest baseline on the recovery-rate and AVM tasks, using identical feature sets and the same out-of-time split. **On this synthetic dataset there is no gradient-boosting accuracy edge worth claiming, and the direction of the tie is not stable.** On the recovery task LightGBM and Ridge are separated by 0.003 (0.533 vs 0.530), with random forest at 0.516; on the v1 book the same comparison had Ridge ahead (0.745 vs 0.727). On the AVM Ridge remains clearly ahead (0.528 vs 0.499, with random forest at 0.464). A difference of 0.003 is not a result in either direction; what the two books agree on is that the recovery DGP is close enough to additive for a linear model to recover it about as well as gradient boosting, which has few nonlinear interactions to exploit. The v2 revision makes the DGP slightly less additive—the collateralisation term is a ratio and the channel introduces a mixture—which is the plausible reason the ordering flipped, and it is a reason to distrust the ordering rather than to claim the new one (erratum item E13).

We report this rather than suppress it: the gradient-boosting and native-categorical machinery the protocol is built around is hypothesised to pay off only on *real* Italian NPL data with genuine nonlinear interactions among judicial district, collateral, and procedure type. On synthetic data, the value of the LightGBM choice is confined to native high-cardinality categorical handling and exact-TreeExplainer SHAP, not to a predictive-accuracy edge.

5.4. Serial-dependence ablation: does purged CV bind?

The purged-CV protocol is the paper’s primary methodological commitment, yet on the baseline synthetic DGP the purge is non-binding by construction (Section 4.1). To avoid leaving the leakage-correction claim merely asserted, we run a controlled ablation that adds the serial dependence the baseline DGP omits and measures whether purging then does anything. Concretely, we inject a vintage-clustered AR(1) latent state $s_t = \rho s_{t-1} + \sqrt{1 - \rho^2} \sigma z_t$ (with $\sigma = 0.10$) over the loans ordered by default date, and let it enter both the recovery target ($y = \text{recovery} + s_t$) and a model-readable covariate `macro_state` = $s_t + \mathcal{N}(0, 0.05^2)$. With $\rho > 0$, temporally adjacent loans share correlated, observable conditions, so a training row that neighbours a test row partially reveals the test row’s state—the look-ahead leakage that purging is designed to sever. The computation reuses the deposited seed-42 loan table and the same recovery-model feature set and hyperparameters (`compute_serial_ablation.py`); every number is reproduced by that script.

Table 6: Baseline Comparison: LightGBM vs Linear and Random-Forest Baselines (Out-of-Time)

Model	Recovery Rate		AVM	
	OOT R^2	OOT MAE	OOT R^2	OOT MAE
Ridge (linear)	0.530	0.109	0.528	28165
Random Forest	0.516	0.112	0.464	30958
LightGBM	0.533	0.110	0.499	29458

Note: Out-of-time evaluation (train pre-2022, test 2022–2025) on the same seed-42 synthetic dataset, identical feature sets and OOT split across models. Categoricals are one-hot encoded for Ridge/Random-Forest and natively handled by LightGBM. The gradient-boosting advantage is task-dependent on this DGP: on the recovery task LightGBM edges Ridge (0.533 vs 0.530); on the AVM task Ridge remains ahead of LightGBM (0.528 vs 0.499). With the v2.1 channel and interaction structure the margins are narrow in both directions; neither model has a material edge on this DGP, and no model-choice conclusion should be read from the ordering; any LightGBM advantage is hypothesised to emerge only on real Italian NPL data with genuine interactions, a claim the v2 real-data release will test. The numbers are synthetic-DGP recoveries, not empirical claims about real Italian NPL portfolios.

Table 7 reports the result, and **on the v2 book it does not support the conclusion v1 drew from it**. At $\rho = 0$ the leakage gap is 0.0006, so the purge is *non-degrading*—that finding survives. But the gaps at $\rho > 0$ run 0.0001–0.0013; they are not ordered in ρ , and only the largest of them exceeds the $\rho = 0$ value. v1 reported a gap that was positive at every $\rho > 0$ and read that as the purge binding once serial dependence is present. That reading is not reproducible here. At this dataset scale every gap—including the one at $\rho = 0$ —is at the noise floor of a fivefold cross-validation on 100,000 rows, and differences of 10^{-3} in R^2 cannot carry the claim.

We therefore withdraw the stronger claim and keep the weaker one. **What this ablation establishes is that purging costs nothing; it does not establish that purging gains anything at this scale.** The mechanism remains plausible and the design remains the right one—the only serial channel here is a single globally-learnable covariate with a 1% embargo, and a separately-verified control in which a purely target-side autocorrelated noise term is made *independent* of all covariates produces no gap at all, confirming that whatever is measured is adjacency-mediated rather than an artefact. But a binding demonstration needs overlapping-horizon labels and the coarser time grids of real Italian NPL panels, which is a real-data question. See erratum item E12.

5.5. κ sensitivity: which numbers are properties of the noise knob

Section 3.3 argues that the headline recovery R^2 is a restatement of the Beta concentration κ rather than a performance result, and that no public aggregate can pin κ down. Rather than defend a value, v2 sweeps it and reports what moves. The grid spans every value anyone has attached a derivation to: 0.77, implied by the noise level of the external benchmark defaulted-loan book; 5 and 12, values this generator has used historically; 20, the deposited

Table 7: Serial-dependence ablation: purged versus standard k -fold cross-validation under an injected autocorrelated latent state (controlled synthetic variant DGP). At this dataset scale the leakage gaps are at the noise floor and do not support a binding claim

Autocorr. ρ	Standard CV R^2	Purged CV R^2	Leakage gap	Std. infl. vs OOT	Purged infl. vs OOT
0.00	0.5851	0.5845	0.0006	-0.0033	-0.0039
0.50	0.5851	0.5846	0.0005	-0.0023	-0.0028
0.90	0.5842	0.5841	0.0001	-0.0038	-0.0039
0.99	0.5875	0.5862	0.0013	0.0013	0.0000

Note: Controlled synthetic ablation, not real-data evidence. We add a vintage-clustered AR(1) latent state $s_t = \rho s_{t-1} + \sqrt{1 - \rho^2} \sigma z_t$ ($\sigma = 0.10$) over loans ordered by default date; the state enters the recovery target ($y = \text{recovery} + s_t$) AND a readable covariate $\text{macro_state} = s_t + \mathcal{N}(0, 0.05^2)$, so temporally adjacent loans share correlated, observable conditions. *Standard CV* is 5-fold on the time-sorted index with no purge/embargo; *Purged CV* is the identical fold geometry with the Lopez de Prado (2018) purge + 1% embargo. The *leakage gap* is Standard $-$ Purged CV R^2 ; *inflation vs OOT* is each CV estimate minus the honest out-of-time R^2 (train pre-2022, test 2022–2025). At $\rho = 0$ (i.i.d. state, the baseline-DGP regime) the purge is non-binding by construction. Only 1 of the 3 gaps at $\rho > 0$ exceeds the $\rho = 0$ gap of 0.0006, and they are not ordered in ρ . **This table therefore does not establish that purging binds.** Every gap here, including the one at $\rho = 0$, is at the noise floor of a fivefold cross-validation on 100,000 rows, and differences of 10^{-3} in R^2 cannot carry that claim. What it does establish is that purging is *non-degrading*: it costs nothing at any ρ tested. The design is not the limitation being reported: the only serial channel here is a single globally-learnable covariate at 100k loans with a 1% embargo, and a separately-verified control in which a purely target-side AR(1) noise term is made *independent* of all covariates produces no gap at all (a true model cannot recall a test row’s noise from its neighbours), so whatever is measured here is adjacency-mediated rather than an artefact. A binding demonstration needs the overlapping-horizon labels and coarser time grids of real Italian NPL panels, which is a real-data question.

constant; 46 and 128, the range whose implied dispersion matches the year-to-year variation of the Banca d’Italia aggregate around its own training mean; and 100 as an upper anchor.

Three readings follow, and the third is the useful one.

The DGP-internal R^2 is κ , restated. Over the valid grid it runs 0.300 to 0.694 with the channel unobserved—most of the range anyone would care about—tracking the closed-form ceiling throughout. A reader comparing the headline against published NPL-model accuracies is comparing against a chosen noise level.

The achieved LTV correlation is κ -bound too. It runs -0.328 to -0.478 across the grid. The -0.422 reported in Section 5.6 therefore holds at $\kappa = 20$ and nowhere else, which is the honest way to state a repair whose magnitude was borrowed from a non-Italian book in the first place (erratum item E2).

The held-out result does not depend on κ at all. Across the entire valid grid the Section 6 level MAE moves from 4.21 to 4.47 pp—a spread of 0.26 pp against the 11.89 pp to 4.42 pp improvement the revision delivers. **This is the cleanest available separation between the one number in this paper that is evidence and the one that is a simulation constant.** It also answers the natural objection to the whole exercise: the out-of-sample claim cannot have been tuned through κ , because κ does not reach it.

A fourth reading is worth one sentence, because it bounds what any version of this paper could claim. The $\kappa = 0.77$ row is the concentration implied by a real defaulted-loan book. Even setting aside the floor defect that makes its R^2 unusable, its closed-form ceiling is 0.1882. A book carrying real defaulted-loan noise admits no headline anywhere near 0.5327,

Table 8: κ sensitivity. The DGP-internal R^2 spans most of its usable range; the held-out result does not move.

κ	Closed-form	Recovery R^2 , OOT (DGP-internal)		Held-out	$\rho(\text{ltv}, \text{recovery})$
	R^2 ceiling	channel unobserved	channel observed	level MAE (pp)	
0.77 [†]	0.1882	-0.0219	-0.0218	10.79	-0.029
5.00	0.4400	0.3001	0.4021	4.21	-0.328
12.00	0.6300	0.4462	0.5969	4.47	-0.391
20.00	0.7333	0.5327	0.7062	4.42	-0.422
46.00	0.8602	0.6330	0.8407	4.41	-0.458
100.00	0.9297	0.6830	0.9080	4.41	-0.477
128.00	0.9441	0.6942	0.9256	4.42	-0.478

Note: $n = 100000$, seed 42, out-of-time split at default year 2022. The closed-form ceiling follows from $\text{Var}(Y | X) = t(1 - t)/(\kappa + 1)$. The held-out level MAE is the Section 6 test re-run at each κ .

[†] **This row does not measure a noise level and must not be quoted as one.** Below $\kappa \approx 1$ the Beta draw’s parameter floors bind on both sides—here for 59.9% of loans—and the draw collapses to Beta(0.5, 0.5), overriding the cell calibration: all four Tav. A2 cells come out between 44.66% and 50.53% instead of 18.87–56.29%. See erratum item E5. The row is shown for completeness and is excluded from the ranges quoted below.

let alone v1’s 0.727.

5.6. SHAP Feature Importance

Table 9 and Figures 1–2 present SHAP feature importance for the AVM and recovery rate models.

For the AVM, region, property type, and macro-zone are the top-3 features, consistent with the hedonic pricing literature [43] in which location dominates property valuation. The condition score and energy class—features calibrated to operational NPL platform parameters—contribute meaningfully, suggesting that granular property characteristics improve valuation accuracy. For recovery rates the top-3 features are collateral_gbv_ratio, region_efficiency and vintage_years.

The recovery ranking changed between v1 and v2, which is the point.. v1’s top three were is_secured, region_efficiency and guarantee_type; is_secured has moved from rank 1 to rank 14. No economics changed. What changed is the generator: with each collateral $\times$ channel cell now recentred on its Banca d’Italia target, the secured/unsecured level lives in the cell offset rather than in a within-cell coefficient, and collateral_gbv_ratio—which in v2 is genuinely pledged share/ltv and is strongly bimodal by secured status—absorbs the role the binary indicator played in v1. **A feature ranking that reorders this much under a revision of the simulator is a description of the simulator.** It is reported here for completeness and for the erratum trail (item E18); no economic inference should be drawn from it, and none is drawn anywhere in this paper.

The two generator defects this table exposed in v1 are now repaired.. v1’s late revisions named two defects visible in this table and did not fix them. v2 fixes both; the before-and-after is in erratum items E1–E3 and is summarised here because it is the clearest illustration of the paper’s central methodological caution.

Table 9: SHAP Feature Importance: Top 20 Features

Rank	AVM Feature	AVM SHAP	RR Feature	RR SHAP
1	region	21418.09	collateral_gbv_ratio	0.0907
2	property_type	12723.48	region_efficiency	0.0333
3	macro_zone	11233.02	vintage_years	0.0263
4	origination_year	5906.38	procedure_type	0.0186
5	energy_score	4365.97	guarantee_type	0.0165
6	superficie_sqm	2220.75	ltv	0.0150
7	judicial_district	1787.94	region	0.0130
8	condition_score	1478.15	condition_score	0.0121
9	debtor_type	1287.98	n_creditors	0.0101
10	region_efficiency	914.99	coverage_ratio	0.0097
11	log_superficie	583.05	judicial_district	0.0075
12	is_secured	442.65	income_gbv_ratio	0.0067
13			years_since_default	0.0056
14			is_secured	0.0042
15			gross_book_value_eur	0.0028
16			collateral_value_eur	0.0026
17			forced_sale_discount	0.0026
18			nbv_gbv_ratio	0.0025
19			debtor_income_eur	0.0024
20			log_income	0.0020

Note: Mean absolute SHAP value for AVM and Recovery Rate models. Features appearing in top-10 of both models indicate consensus drivers. Proprietary-model-calibrated features (region_efficiency, condition_score, forced_sale_discount) among top contributors.

First, `ltv` had no causal path to recovery. In v1 collateral value was *defined* as property value $\times$ `ltv` and GBV as collateral value $\times (1 + \text{accrued})$, so the derived collateral-to-GBV ratio reduced to $1/(1 + \text{accrued})$ and was independent of `ltv` by construction ($\rho = -0.0036$). `ltv` therefore correlated with generated recovery at -0.034 and appeared in this table at rank 17 with importance 0.0020—a faithful report of the generator, and the reverse of the regularity observed on real defaulted-loan data. On an external benchmark loss-given-default file of 2545 defaulted loans—`lgd.csv`, the companion dataset of Baesens et al. [6]—LTV correlates with recovery at -0.424 and is the stronger of the two genuine covariates that file provides. In v2 the loan is the primitive: GBV is property value $\times$ `ltv` and collateral is property value $\times$ the realisable share, so the ratio moves with `ltv`. The measured correlation is now -0.422 and `ltv` enters this table at rank 6 with importance 0.0150. We remain explicit about what the benchmark comparison is and is not: it is a general retail/mortgage LGD file, *not* an Italian corporate NPL book, and we make no claim that its coefficient transfers to the Italian secured-NPL setting—only that LTV carries substantial recovery signal on real defaulted loans against essentially none in the v1 generator. Because the companion website states no redistribution licence, the file is not copied into the replication package but is pinned by SHA-256 in a fetch-and-verify script (`code/data/external/fetch_lgd.py`; Appendix [Appendix A](#)). By the same repair `collateral_gbv_ratio` is no longer misnamed, and the accrued-interest quantity it used to encode is exported as `accrued_interest_ratio`.

Second, the generated LTV distribution excluded both tails that matter. v1 clipped `ltv` to $[0.40, 1.0]$ with realised support $[0.559, 1.000]$: *no under-collateralised loans at all*—the segment where recovery risk concentrates—and no well-collateralised ones either, at roughly a quarter of the real dispersion. v2’s support is $[0.059, 3.000]$ with mean 0.677, SD 0.363, 15.3% above LTV 1.0 and 36.4% below 0.5, against 17.5% and 34.5% in the benchmark file. **The magnitude of this repair is borrowed and is not Italian.** Banca d’Italia Note 48 has no LTV dimension, so the target correlation and moments come from a non-Italian book and are a free parameter in the same sense κ is; Section [5.5](#) shows the achieved correlation is κ -bound in any case. What is *not* borrowed is the structural half of the fix: correcting the collateral definition alone, with the explicit LTV term switched off, moves the correlation from -0.034 to -0.332 .

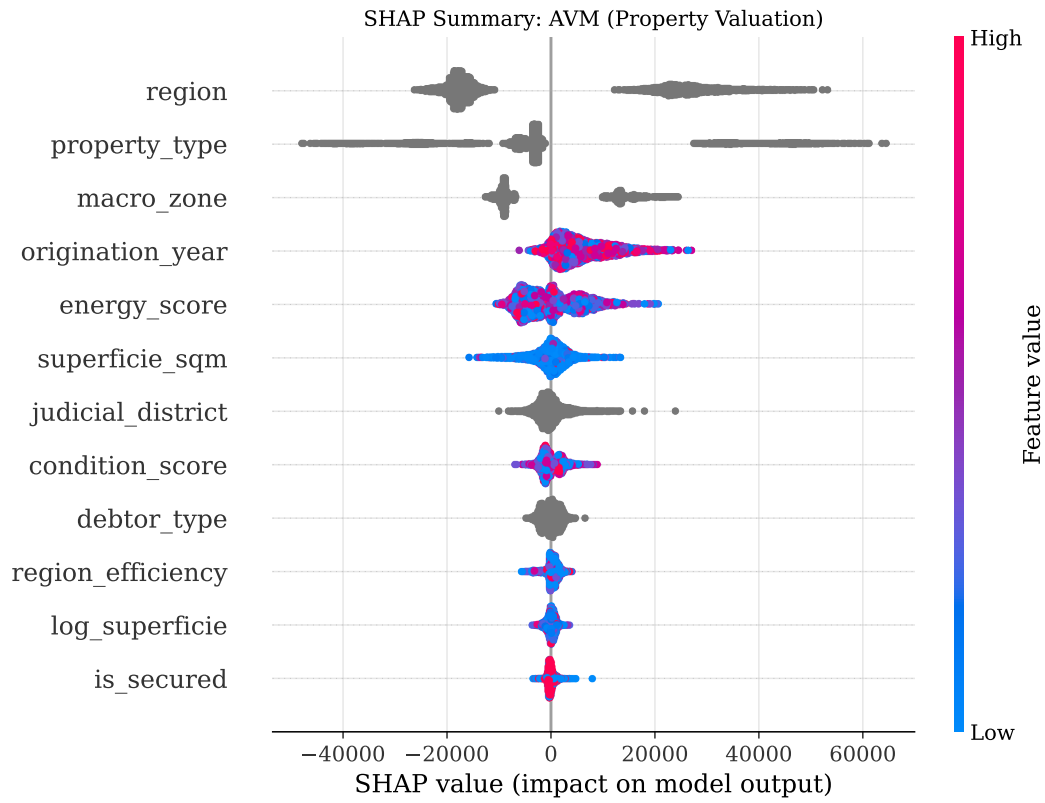


Figure 1: SHAP Summary Plot: AVM Model

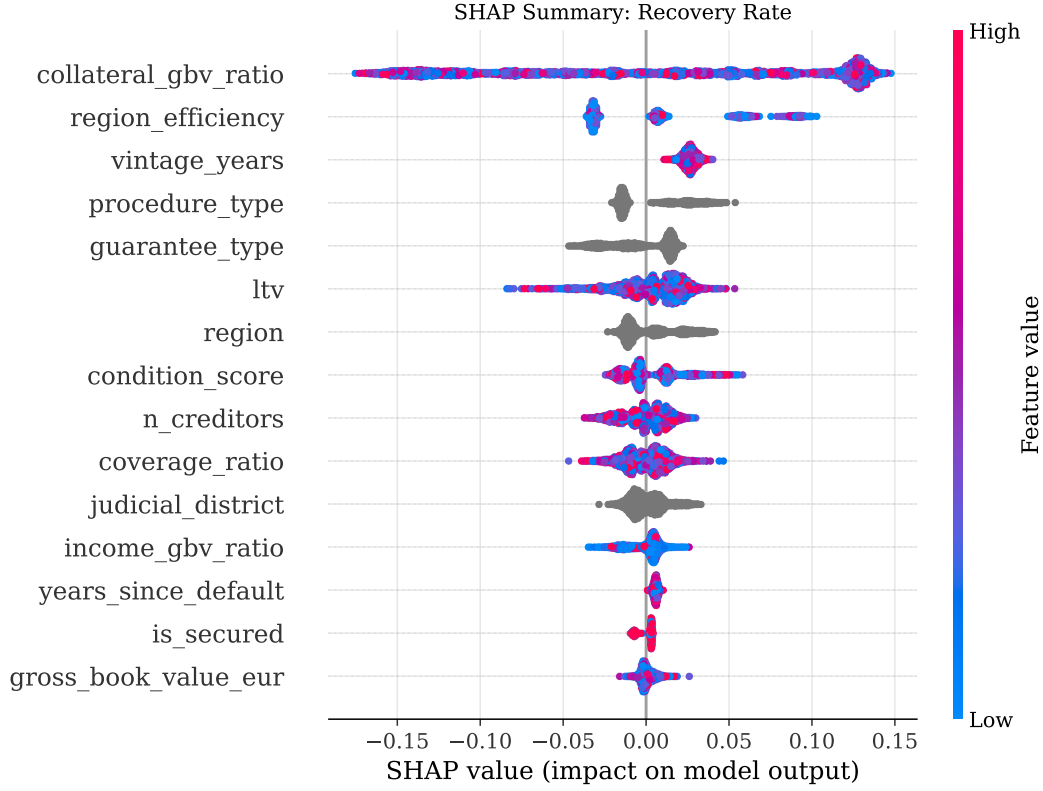


Figure 2: SHAP Summary Plot: Recovery Rate Model

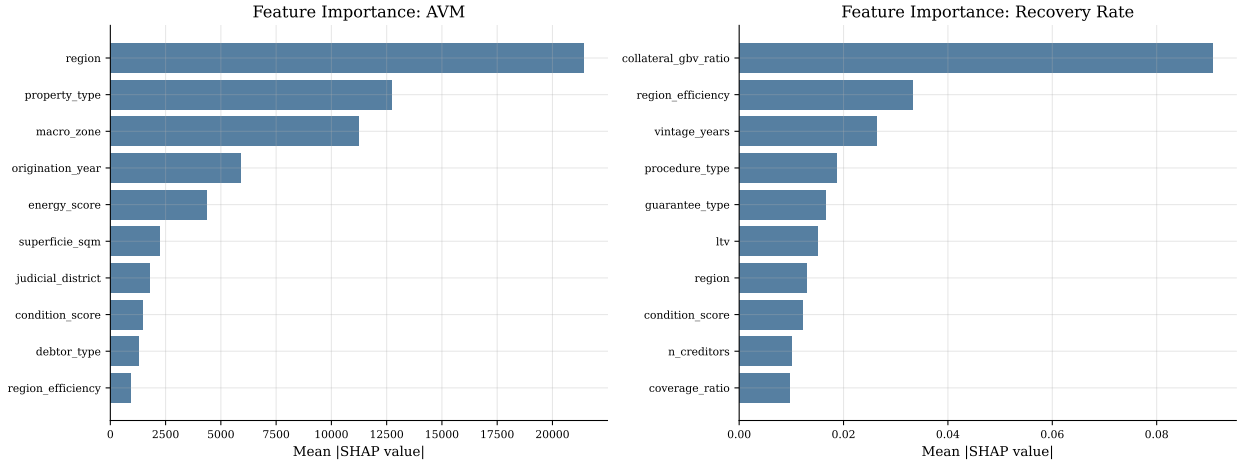


Figure 3: Feature Importance Comparison: AVM vs Recovery Rate

5.7. Weibull Recovery Curves

v1's round-trip is withdrawn.. v1 presented, as “the one genuinely external-aggregate validity check the synthetic paper can offer”, the recovery of its own hardcoded per-district medians (Milano 24, Roma 48, Napoli 72, Calabria 84+ months) to within 4.66% mean absolute error over 26 districts. Two things are wrong with that. Those durations do not correspond to

anything in the current Ministero della Giustizia public release and do not reproduce on 2025 court data (erratum item E9); and recovering a number one has injected is self-consistency, not external validity, however small the error. Scored the same way, the v2 book misses those same stale targets by 61.76%—by construction, because it is no longer calibrated to them. The table has been removed rather than reprinted with a caveat. Section 6 is the external check v1 lacked.

What the v2 curves show.. Figure 4 displays the estimated Weibull recovery curves by segment. Fitted medians by macro-zone are 14.3 months North, 16.1 Centre and 17.2 South, a South/North ratio of 1.207 (fitted shapes 1.028 and 1.014, i.e. close to exponential). v1 reported 26.6 and 65.1 months for a ratio of 2.44—roughly double. Conditioning matters here and the two ratios in this paper are *not* the same statistic: 1.207 pools both closure channels, while the 1.28 quoted in Section 6 is for worked-out positions only, where the judicial gradient actually applies. Pooling dilutes the gradient because a sold position closes on its transfer date irrespective of the court.

These remain calibration quantities, not measurements. The level is the Tav. 4 cohort curve the generator was anchored to and the gradient is the SIECIC multiplier it was given; recovering them is expected. What is *not* circular is that the same generator, scored on cohorts it never saw, misses by 6.92 pp rather than v1’s 42.86 pp (Section 6).

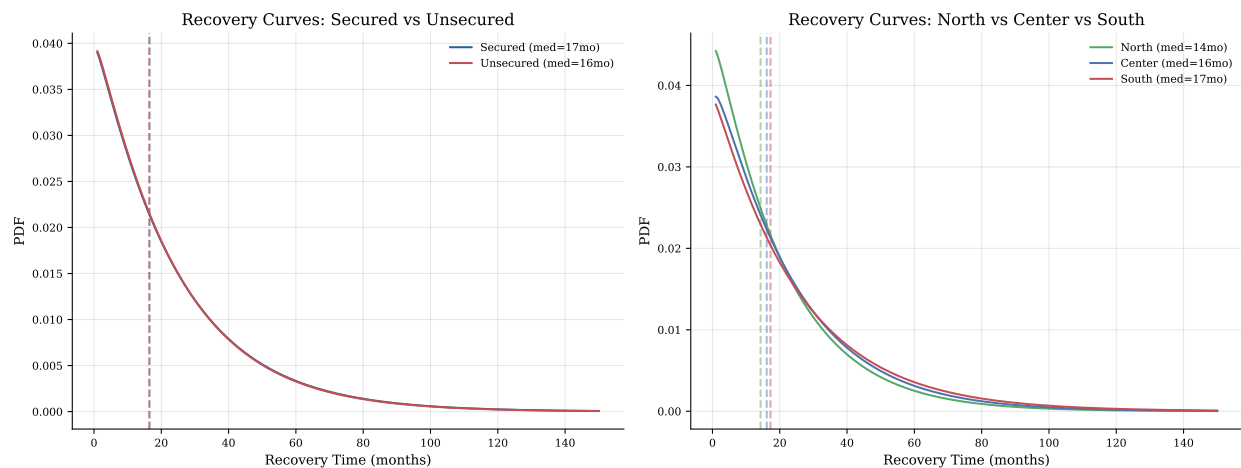


Figure 4: Weibull Recovery Curves by Segment

5.8. Regional Recovery Heterogeneity

Figure 5 visualizes mean recovery rates across Italian regions. On the v2 book the North averages 59.3% and the South and Islands 40.9%, ranging from Piemonte at 63.0% to Campania at 39.8%. v1 reported Northern regions above 40% and Southern below 25%; both levels moved once the cells were calibrated to Banca d’Italia rather than to 0.45/0.15 (erratum item E19). The gradient mirrors the judicial duration pattern, and region_efficiency is the second-ranked recovery driver in the SHAP attribution of Table 9. **This figure has no external validation of any kind.** Banca d’Italia Note 48 carries no territorial dimension, so the recovery side of the North–South story is exactly as untested in v2 as it was in v1; SIECIC constrains only the timing gradient. The map displays an injected calibration input.

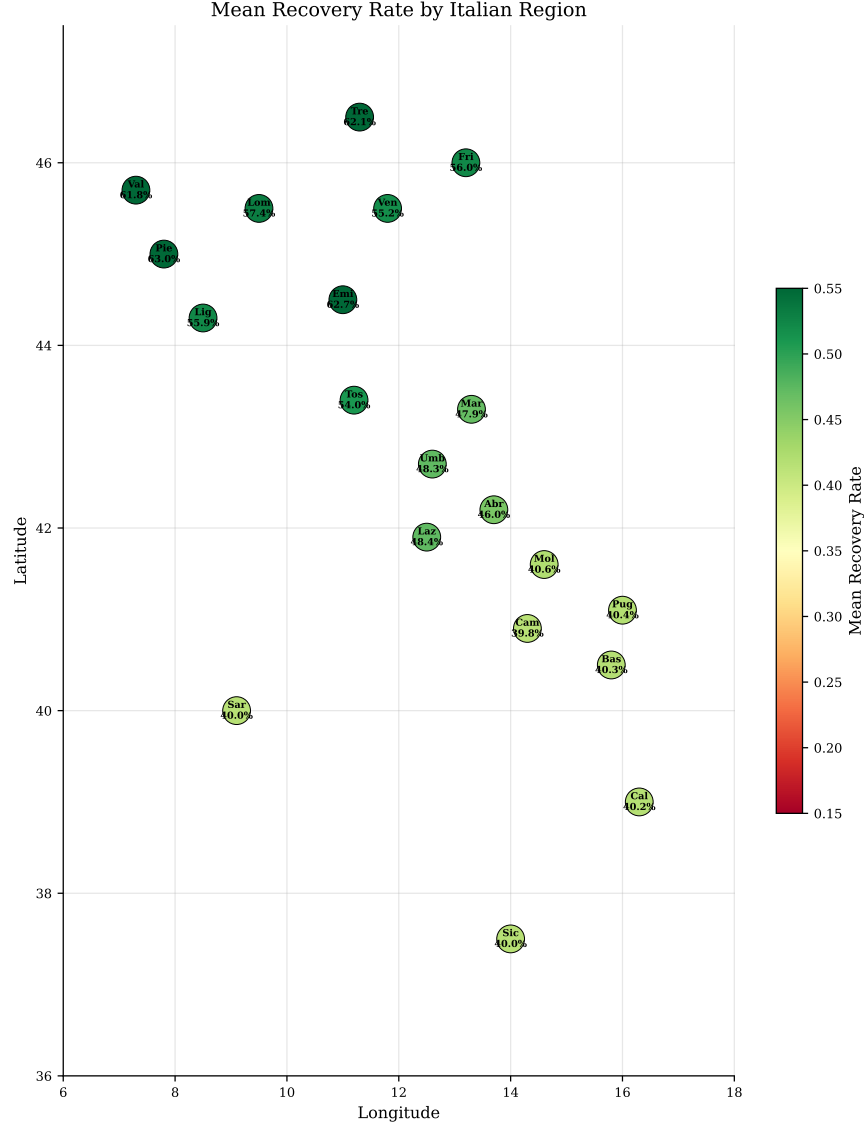


Figure 5: Mean Recovery Rate by Italian Region

6. Held-out anchoring against supervisory data

This section is new in v2 and is the paper’s only non-circular quantitative result. Everything in Section 5 measures how well the protocol recovers a process written down in this repository. What follows measures whether the process itself predicts supervisory data it was not shown.

6.1. Design

The test rests on one property of the generator described in Section 3: its Banca d’Italia constants are computed on exit years 2006–2019 and entry cohorts up to 2016 only. Two held-out sets follow.

Recovery levels. Tav. A2 gives the recovery rate for each of the four collateral $\times$ channel cells, by exit year. The generator’s realised cell means are scored against exit years 2020–2024: 20 predictions.

Closure timing. Tav. 4 gives the cumulative share of positions closed within 1...6 years, by entry cohort. The generator’s realised closure curve is scored against cohorts 2017–2023: 27 testable points.

Neither set was available to any constant in the generator, so a miss is a real miss. Two further design points matter for reading the result. First, v1 is not reconstructed or approximated: it is extracted from git revision `7e42680` and executed, so the “before” column is the deposited code. Second, v1 has no channel dimension, so it necessarily predicts the same rate for the worked-out and the sold cell of a given collateral class. That is the defect being measured and it is scored as such rather than excused.

Both books are generated at $n = 100000$, seed 42, and $\kappa = 20$.

6.2. Result

Table 10: Held-out scores against Banca d’Italia Note 48. Neither the exit years nor the entry cohorts below were visible to either generator’s calibration.

	v1 as deposited	v2	Benchmark
<i>Recovery levels</i> — Tav. A2, 20 predictions			
MAE (pp)	11.89	4.42	4.43
Bias (pp)	5.90	-0.53	—
Max abs. error (pp)	31.18	11.28	—
Models the closure channel	no	yes	—
<i>Closure timing</i> — Tav. 4, 27 predictions			
MAE (pp)	42.86	6.92	7.40
Bias (pp)	42.86	6.77	—
Max abs. error (pp)	57.47	24.07	—
Points under-predicting	27/27	25/27	—
<i>Judicial gradient</i> — median closure time, worked-out positions			
South / North ratio	2.52	1.28	1.325

Note: The level benchmark is the best calibration rule the source admits—a flat per-cell mean fitted on the same training window—so 4.42 pp is at the ceiling this data permits, not merely better than v1. The timing benchmark is the best train-only rule of the same design (carry the last training cohort forward). The gradient benchmark is the South/North ratio of CEPEJ disposition time measured across 132 tribunals in the 2025 SIECIC release; v1’s own text reported 2.44. Positive bias means the generator under-predicts.

Levels: a pass, at the ceiling, with the bias cancelling. The improvement from 11.89 pp to 4.42 pp is almost entirely the two cells v1 could not represent. Table 11 breaks it out. The near-zero pooled bias (-0.53 pp) must not be quoted alone: two cells miss systematically in

opposite directions and cancel. The secured worked-out cell is now the worst of the four at 7.03 pp with all 5 years signed the same way—not a new defect, but the same out-of-sample drift the benchmark rule exhibits.

Table 11: Held-out recovery levels by cell, exit years 2020–2024 (percent, and percentage points)

Cell	v1 predicts	v2 predicts	Realised	v1 MAE	v2 MAE
Secured, worked out	48.74	56.33	49.30	1.12	7.03
Secured, sold	48.74	38.47	37.03	11.71	2.37
Unsecured, worked out	14.40	38.20	39.38	24.98	3.11
Unsecured, sold	14.40	19.00	24.17	9.78	5.17

Note: v1 predicts the same value twice within each collateral class because it has no channel dimension. Its secured worked-out cell was in fact close (1.12 pp); its error is concentrated in the unsecured worked-out cell (24.98 pp), which is the 0.15 unsecured target of erratum item E7 priced against a realised 39.38%.

Timing: no longer refuted, and still biased in one direction. v1’s closure curve implied that 9.57% of positions close within one year; the held-out cohorts realised 42.0–64.8%. v2’s curve gives 40.72%. The MAE falls from 42.86 pp to 6.92 pp, slightly beating the best train-only rule (7.40 pp). **This is an improvement, not a vindication.** 25 of 27 points still under-predict and the residual bias is 6.77 pp, the same sign as v1’s 42.86 pp. The generator is anchored on cohort 2016 and the cohorts kept accelerating after it; the worst single point is cohort 2021 at 1 year, predicted 40.72% against a realised 64.78%.

The anchor choice is documented rather than hidden. Three train-only calibrations were available—carry the last training cohort forward, pool the training cohorts, or extrapolate their fitted trend—scoring 7.40, 25.83 and 11.57 pp respectively on the held-out cohorts. v2 uses the first. The train-only argument for it is that the training cohorts alone show monotone acceleration whose main driver, the securitisation wave, had largely arrived by 2016, so projecting it further is unjustified. **We state plainly that this is not a blind pre-registration:** the three held-out scores were known when the choice was made. All three variants ship, and a reader who prefers the trend variant will get a worse number.

Both structural contrasts are generated too wide. The generator gets the levels right and the *contrasts* too large, in both dimensions. The collateral spread on worked-out positions is generated at 18.13 pp against a realised 9.93 pp (a shrinkage factor of 0.548); the channel margins are generated at 17.86 and 19.19 pp against realised 12.27 and 15.20 pp. v1’s collateral spread was worse still (34.34 pp, shrinkage 0.289), but the direction of the error is unchanged and it is not repaired by this revision.

6.3. What this does and does not establish

Establishes. The generator’s recovery *marginals* are out-of-sample credible against supervisory data on a genuine temporal split, at the accuracy ceiling the source permits. The timing layer is no longer inconsistent with the only published timing statistic for its own target population. The judicial gradient is now 1.28 against the 1.325 measured on current court data, a shortfall of 3.0%, against v1’s 2.52 which missed the same benchmark

in the opposite direction and by thirty times as much (signed shortfall -90.1%, i.e. a large overshoot). Note the direction of the residual miss: v2 generates this gradient slightly *too flat*, not too steep.

Does not establish. Nothing about the loan-level results. Banca d’Italia publishes no loan-level data, so no test built on this source can validate a loan-level R^2 ; the $R^2 = 0.533$ of Section 5 still measures DGP recovery. The generator’s *structure* is untested—only collateral, channel and closure timing appear anywhere in the annex, and the recentring in Section 3 makes the calibration-window level a round-trip by construction. The geography story gets no recovery-side support at all, because Note 48 has no territorial dimension; SIECIC constrains the timing gradient only. And the timing layer, though six times closer, is still biased in one direction.

7. Erratum: corrections to v1

Every item below changes a number or a claim in the deposited v1. The “was” column is read out of git revision 7e42680 rather than retyped, so each pair is provably the deposited value. Items E1–E4 were *disclosed* in late revisions of v1 but not repaired; v2 repairs them.

- E1. **ltv had no causal path to recovery, and now has one.** v1 built ltv, exported it, and included it among the 24 recovery-model features, but it entered no component of the recovery signal: collateral was *defined* as property value $\times$ ltv and GBV as collateral $\times$ (1+accrued), so the collateral-to-GBV ratio reduced to $1/(1+\text{accrued})$ and ltv cancelled algebraically (measured $\rho = -0.0036$). Consequently $\rho(\text{ltv}, \text{recovery}) = -0.034$ and ltv appeared at SHAP rank 17 with importance 0.0020. v2 makes the loan the primitive: GBV is property value $\times$ ltv, and collateral is property value $\times$ the share the lender can actually reach, so their ratio moves with ltv as it should. Now $\rho(\text{ltv}, \text{recovery}) = -0.422$, $\rho(\text{collateral}/\text{GBV}, \text{ltv}) = -0.566$, and ltv ranks 3 of 24 by model gain. *Most of this repair required no borrowed parameter:* correcting the collateral definition alone moves the correlation from -0.034 to -0.332, and the explicit LTV term carries only the remainder.
- E2. **The generated book contained no under-collateralised loan.** v1 clipped ltv to [0.40, 1.00] with a realised support of [0.559, 1.000]—no loan above LTV 1 and none below 0.5, i.e. neither tail that matters. v2’s support is [0.059, 3.000] with 15.3% above 1 and 36.4% below 0.5, against 17.5% and 34.5% in the external reference book. **Provenance caveat, stated plainly:** the LTV moments and the target correlation -0.424 come from the Baesens et al. [6] loss-given-default file ($n = 2545$), which is *not* an Italian book. Banca d’Italia Note 48 has no LTV dimension and the national OMI bulk file is behind an Italian digital identity, so no Italian public source on disk carries loan-level collateralisation for defaulted exposures. The sign is a domain regularity and the widened support is a structural requirement; the *magnitude* is a free parameter in exactly the sense κ is, and Section 5.5 shows it is κ -bound in any case.
- E3. **collateral_gbv_ratio was misnamed.** In v1 it reduced to $1/(1 + \text{accrued})$ and measured accrued interest, not collateralisation. In v2 it is genuinely collateral over

exposure; the accrued-interest quantity it used to encode is exported separately as `accrued_interest_ratio`.

- E4. **κ was a hard-coded constant and is now an argument**, with a sensitivity sweep (Section 5.5) rather than a single quoted value. The default is unchanged at 20 for comparability, not because it is defended.
- E5. **New defect found while sweeping κ , now guarded**. The Beta draw floors both shape parameters at 0.5. Below $\kappa \approx 1$ both floors bind at once and the draw collapses to $\text{Beta}(0.5, 0.5)$ —mean 0.5 regardless of the target—silently overriding the cell calibration. At $\kappa = 0.77$ this affects 59.9% of loans and all four Tav. A2 cells collapse into 44.66–50.53% instead of 18.87–56.29%. The generator now warns and records the affected share, and low- κ rows are excluded from every reported range.
- E6. **The generator had no closure-channel dimension**. v1 modelled a single recovery process, so it could not represent the sale channel at all. Earlier drafts of this item said that channel “carries roughly half of closures by amount”; that figure is an artefact of amount-weighted pooling, which this paper elsewhere declines to use (Table 2), and it is corrected here. Over the calibration window the sold share is 0.294 by amount and 0.318 by number under equal-year weighting, 57.1% and 42.0% pooled, and it ranges from 2.6% of closed amounts in 2006 to 86.2% in 2018. No single share summarises the channel, and the argument for modelling it does not need one. v2 adds the dimension, calibrated to Tav. A1 and Tav. A2. Note that this is the change that *lowers* the headline R^2 (Section 5).
- E7. **The recovery targets 0.45 secured and 0.15 unsecured appear in no Banca d’Italia table**. The second is the larger error. Banca d’Italia’s unsecured *worked-out* recovery rate averages 38.27% over the calibration window, not 15%; against the held-out years v1’s unsecured cells missed by 24.98 and 9.78 pp. v2 targets the four published cells directly.
- E8. **The closure-timing calibration was refuted, twice, and is re-anchored**. v1 drew Weibull shape 1.5 around a national median of 54 months, implying 7.0% of positions close within a year against a realised 42.0–64.8%. Against the held-out Tav. 4 cohorts it missed by 42.86 pp with every point in the same direction. This confirms, from a second and independent source, the negative finding already obtained from 2025 court data. v2 anchors to the Tav. 4 cohort-2016 curve (shape 1.0219, median 16.5 months) and misses by 6.92 pp—improved, not fixed.
- E9. **The per-district durations Milano 24 / Roma 48 / Napoli 72 / Calabria 84+ months do not reproduce, and their citation does not resolve**. No such figures appear in the current Ministero della Giustizia public monitoring release. On 2025 SIECIC data the four named courts sit in a 15–17 month band and the South/North median ratio across 132 tribunals is 1.325, not the 2.44 v1 reported. v2 replaces the absolute medians with a relative disposition-time multiplier and obtains 1.28. *Caveat*:

disposition time is a stock/flow proxy and understates duration while disposals accelerate, so this does not show the old figures were wrong for their original period—it shows they are stale and no longer reproduce in level or in ordering.

- E10. **The “4.7% mean absolute error” Weibull round-trip is withdrawn as a validity check.** v1 presented the recovery of its own hardcoded district medians (4.66% mean absolute error over 26 districts) as “the one genuinely external-aggregate validity check the synthetic v1 paper can offer”. It was a round-trip against a calibration input that item E9 shows to be stale, so it measured self-consistency, not external validity. Scored the same way, the v2 book misses those same stale targets by 61.76%—by design. Section 6 is the external check that v1 lacked.
- E11. **The North–South Weibull medians change.** v1 reported fitted medians of 26.6 months North against 65.1 South, a ratio of 2.44. On the v2 book the medians are 13.0 and 16.7 months, a ratio of 1.28. The gradient survives; its magnitude was roughly double what current court data support.
- E12. **The serial-dependence ablation no longer supports the claim that the purge binds.** On the v1 book the leakage gap was positive at every $\rho > 0$ and v1 read that as the purge binding once serial dependence is present. On the v2 book the gap at $\rho = 0$ is 0.0006 and the gaps at $\rho > 0$ run 0.0001–0.0013; they are not ordered in ρ and only the largest exceeds the $\rho = 0$ value. At this dataset scale every gap is at the noise floor. The surviving claim is the weaker one: the purge is *non-degrading*. Section 5.4 is rewritten accordingly.
- E13. **The baseline ordering changes on the recovery task.** v1 reported Ridge ahead of LightGBM (0.745 vs 0.727). On the v2 book they are 0.530 and 0.533, a gap of 0.003. The correct reading is unchanged—the two are indistinguishable and no gradient-boosting accuracy claim is warranted—but the direction of the tie has flipped and v1’s wording should not be repeated. On the AVM Ridge remains ahead (0.528 vs 0.499).
- E14. **The train-on-synthetic, test-on-real transfer still fails, but v1’s explanation of why is withdrawn.** Re-run on the v2.1 book, the single-covariate transfer to the 2,545-loan `lgd.csv` file scores $R^2 = -0.767$ (LightGBM) and -0.754 (Ridge), against v1’s -1.12 and -1.20 . The verdict—worse than predicting the real sample mean—is unchanged. The diagnosis is not. v1 attributed the failure to support mismatch (57.2% of real loans outside the synthetic `ltv` support, KS 0.503) and to near-constant predictions arising from an `ltv` that carried no signal ($R^2 = 0.0009$ within the synthetic book, prediction SD 0.026). The `ltv` repair of item E1 removed both: only 1.65% now falls outside the support (KS 0.0919), restricting to the in-support 98.3% moves the score to -0.766 , and `ltv` now predicts recovery within the synthetic book at $R^2 = 0.1848$. What remains is a level mismatch—synthetic recovery averages 0.459 against 0.772 on a differently-calibrated population—while the ordering transfers (predictions correlate with realised recovery at 0.400). Section 8.4 is rewritten accordingly. This is the item to read as a caution about *our own* negative results: v1’s causal story for a failure was as revision-dependent as the SHAP ranking in item E18.

- E15. **The SIECIC South/North benchmark was quoted as 1.22 and is 1.325; the direction of v2’s residual miss reverses.** Five places in earlier drafts of this revision compared the generated judicial gradient against a benchmark of 1.22 “measured on 2025 court data”. That figure appears in no machine-readable output of this project; the generator’s own calibration record gives the South/North median ratio of CEPEJ disposition time over the same 132 tribunals in the same release as 1.325, and Table 1 has been scoring the book against 1.325 throughout. Against the corrected benchmark the v2 gradient of 1.28 *under-shoots* by 3.0% rather than overshooting, so the sign of the residual error in the manuscript’s own text was wrong. Every comparison now reads the macro. This defect is the same species as item E16—a hand-typed number surviving next to a generated one that disagreed with it—and it is why the discipline of this revision is that no statistic is typed by hand.
- E16. **Seven included tables were v1’s, and the manuscript’s own text contradicted them.** The tables directory carried the pre-remediation copies of `tab_model_performance`, `tab_oot_validation`, `tab_baselines`, `tab_shap_top20`, `tab_serial_ablation`, `tab_calibration_moments` and `tab_injected_vs_learned`, so the printed tables showed v1’s recovery R^2 of 0.746/0.727, v1’s SHAP ordering, and v1’s 0.45/0.15 recovery targets while the surrounding text stated the v2 values. All seven are regenerated from the v2.1 pipeline in this revision. Three of them asserted, in their notes, claims this paper formally retracts elsewhere—that the purge binds, that Ridge leads on both tasks, that the reported recovery R^2 is 0.73—for `tab_serial_ablation` and `tab_injected_vs_learned` those notes are now derived from the computed numbers; in the post-verification residual round `tab_baselines`’ verdict sentence was likewise converted to a derived comparative and `tab_model_performance`’s schema counts were corrected at the emitter. `tab_injected_vs_learned` had no producer at all and could not be regenerated; it has one now (`compute_injected_vs_learned.py`), which fails if the generator’s component set drifts from the tabulated one.
- E17. **Disclosed, not fixed: the book’s secured share is wrong by a factor of four.** The generated book is 70.6% secured by count; Banca d’Italia’s closed positions are 16.0% secured by number (56.8% by amount). Correcting this would move every book-level moment in the paper and it lies outside the scope of this revision, so it is recorded rather than patched. The conditional cell calibration is unaffected, which is why the held-out test of Section 6 still passes.
- E18. **The recovery-model SHAP top-3 changes completely.** v1: `is_secured`, `region_efficiency`, `guarantee_type`. v2: `collateral_gbv_ratio`, `region_efficiency`, `vintage_years`. `is_secured` falls from rank 1 to 14 and `ltv` rises from 17 to 6. Nothing economic changed; the cell recentring moved the secured/unsecured level out of the within-cell signal and the corrected collateralisation ratio absorbed it. Any statement of the form “the protocol finds X to be the leading recovery driver” should be read as a statement about the simulator, in v1 and in v2 alike.
- E19. **Regional recovery levels change.** v1’s regional map claimed Northern regions above

40% and Southern below 25%. On the v2 book, calibrated to the Banca d’Italia cells, the North averages 59.3% and the South 40.9%, ranging from Piemonte at 63.0% to Campania at 39.8%. The gradient survives; the levels were understated because v1’s 0.15 unsecured target (item E7) pulled the whole book down.

- E20. **Schema counts change.** The loan table grows from 37 to 40 columns and from 31 to 34 modelling-eligible features, with the addition of `closure_channel`, `pledged_share` and `accrued_interest_ratio`. The recovery model still uses 24 features and the AVM 12. `closure_channel` is deliberately *not* among the recovery model’s features (Section 5).

8. Discussion

8.1. Implications (conditional on real-data validation)

The following are *hypotheses the protocol is designed to test on real data*, not findings established by this synthetic study. Because the synthetic SHAP attribution and the North–South ratio recover injected calibration inputs (Sections 3.3 and 5.7), none of the points below is licensed by the synthetic results alone; each becomes a claim only after the pre-registered v2 real-data evaluation (Section 8.4).

For regulators.. On synthetic data the protocol assigns regional judicial efficiency a high SHAP rank (second, after `collateral_gbv_ratio`), which is by construction—it recovers the judicial-efficiency component injected into the DGP. Two qualifications belong with that sentence rather than in a footnote. The feature it now ranks behind is not the one v1 reported: `is_secured` led the v1 table and sits at rank 14 on the v2 book, so the ordinal claim is a property of the simulator revision (erratum item E18). And the judicial component is calibrated to a *relative* SIECIC disposition-time gradient, not to published absolute district durations—those do not reproduce in the current release (erratum item E9). The result is consistent with, but does not independently establish, the well-documented hypothesis [20] that judicial efficiency is a leading determinant of Italian NPL recovery; whether the protocol surfaces judicial efficiency as a dominant driver on *real* loan-level data is a question for v2.

For NPL investors.. The framework’s loan-level predictions, combined with the illustrative ZOPA rule, are intended to provide a starting point for portfolio bidding once validated on real data. We make no claim that the synthetic out-of-time estimates are realistic for deployment; the headline metrics are scaffold-grade pending v2.

For banks.. The serial-dependence ablation (Section 5.4) shows purged cross-validation is non-degrading on synthetic data; it does *not* show that purging binds, and this revision withdraws that claim (erratum item E12). The expectation that purging produces more conservative—and more honest—estimates than standard k -fold on real serially-dependent panels therefore rests on the mechanism and on Lopez de Prado [31], not on anything measured here. Banks using standard CV for internal model validation on serially-dependent data may over-report model accuracy, a concern the protocol is designed to address but that this synthetic study cannot quantify at real-panel magnitude.

8.2. Comparison with Basel IRB

The Basel IRB framework [40] requires banks to estimate probability of default (PD) and loss given default (LGD) for capital calculation. Our recovery rate model directly estimates $1 - \text{LGD}$, and the Weibull curves provide the “downturn LGD” distribution that Basel III requires. The SHAP explanations satisfy the interpretability requirements that supervisors impose on internal models [8].

8.3. Limitations

What v2’s held-out test does and does not relax.. The limitations below are v1’s, and Section 6 relaxes exactly one of them: the recovery layer’s *marginals* are no longer a round-trip. It does not touch the loan-level results, the DGP structure, or the geography story, and it leaves the timing layer improved but still one-sided. Three limitations are also *added* by this revision and are stated here rather than left to the erratum: the serial-dependence ablation no longer supports the claim that purging binds (item E12); the book’s secured share is wrong by a factor of four and was not fixed (item E17); and the LTV magnitude that repairs item E1 is borrowed from a non-Italian book, so the one feature v1 got structurally wrong is now structurally right and quantitatively unanchored.

Several limitations warrant discussion. *First*, the use of synthetic data means that our results are illustrative rather than empirical. While the calibration to aggregate statistics and operational platform parameters produces plausible distributions with realistic feature correlations, real loan-level data would contain patterns that synthetic generation cannot capture (e.g., servicer-specific workout strategies, loan-level renegotiation histories, and actual appraisal reports). The framework should therefore be evaluated as a *methodology contribution* demonstrating ML best practices for NPL valuation, not as empirical findings about actual Italian NPL portfolios. *Second*, the enriched feature set (the 40-column loan table, of which 34 are modelling-eligible features) is richer than a minimal baseline specification, but real data would include additional granular variables such as detailed borrower financials, workout activity logs, and auction-level pricing histories; we do not claim a predictive edge for the enriched set on synthetic data (see the baseline comparison below). *Third*, the unsecured credit model achieves a modest DGP-internal OOT $R^2 = 0.219$ on the v2 book (v1 reported 0.339), indicating that unsecured recovery prediction remains inherently more challenging due to the absence of collateral signal; additional debtor-level features (e.g., employment status, asset declarations) would likely improve performance. *Fourth*, the executed baseline comparison (Table 6, Section 5.3) shows that on this additive synthetic DGP LightGBM does not beat a Ridge linear baseline; we therefore make *no* accuracy claim for gradient boosting on synthetic data and motivate the LightGBM choice solely by its native high-cardinality categorical handling and exact-TreeExplainer SHAP. Modern tabular deep-learning architectures (TabNet [2], FT-Transformer [22]) are a natural further comparison, but only on real Italian NPL data with genuine nonlinear interactions; on the additive synthetic DGP they would, like LightGBM, have no interaction structure to exploit. We treat that comparison, and any claim of a gradient-boosting or deep-learning accuracy edge, as a v2 follow-up requiring real-data access.

8.4. External-aggregate validation: executed, and partially

v1 deferred external-aggregate validation to this release on the grounds that it required source-document loaders that were not implemented. Half of it is now executed and reported in Section 6: the Banca d'Italia leg, against *Note di stabilità finanziaria e di vigilanza* N. 48 rather than the Bollettino Statistico, on a genuine temporal split.

Two qualifications on what that discharges. First, it validates the *generator's marginals*, not the synthetic-trained *model*—and it could not do otherwise, because Note 48 publishes no loan-level data and a loan-level R^2 has no aggregate counterpart. Second, the GACS leg is not executed. Deal-level servicer reports for the individual securitisation SPVs are public investor-reporting documents and do contain the actual-versus-business-plan collection curves this paper wants, but obtaining them is a per-deal scrape rather than a single download, and it was scoped out of this revision rather than attempted and abandoned. It remains the strongest untried route to an external check on the *timing* layer, which is the layer Section 6 leaves biased.

8.5. Path to real-data validation

A real-data follow-up requires partnership with a major Italian NPL servicer (e.g., doValue, Cerved, Prelios) or with one of the GACS-securitization vehicles whose performance data is available to investors. The framework is designed to be drop-in: once a real loan-level table is available, the same purged-CV pipeline, LightGBM categorical handling, and SHAP attribution apply without modification. The replication package includes a schema-translator that maps between the synthetic schema used here and the typical servicer schema. Two follow-up evaluations would close the principal evidential gap: (i) train-on-synthetic, evaluate-on-real to test the framework's transferability; and (ii) train-on-real, compare-to-incumbent-IRB to quantify the regulatory-capital implications of switching from logistic-regression-based PD/LGD models to gradient-boosted alternatives.

Track (i) is executed here, and it fails—for a different reason than v1 reported (re-run on the v2 book, 2026-08-18).. Track (i) does not in fact require servicer access, because this paper already pins a real defaulted-loan file by SHA-256: the `lgd.csv` benchmark of Section 5.6. We therefore run it rather than defer it (`code/data/external/transferability_check.py`). Since `ltv` is the only covariate common to the two schemas, this is a *single-covariate* transfer and so an upper bound on what a shared-feature transfer could achieve, not a replication of the 24-feature pipeline on real data. A recovery model trained on the synthetic book and applied to the 2,545 real defaulted loans attains $R^2 = -0.767$ (LightGBM) and -0.754 (Ridge)—*worse than predicting the real sample mean*, and only modestly better than the -0.906 obtained by predicting the synthetic mean as a constant. **The transfer still fails; the diagnosis has changed.**

v1 attributed the failure to support mismatch and to near-constant predictions. Neither survives on the v2 book, because the `ltv` repair of erratum item E1 removed both. Only 1.65% of real loans now fall outside the synthetic `ltv` support (two-sample KS = 0.0919, against v1's 57.2% and 0.503), and restricting to the 2,503 in-support loans—98.3% of the file—moves the score to -0.766 , i.e. not at all. Within the synthetic book `ltv` now predicts recovery at $R^2 = 0.1848$ (v1: 0.0009), and the two books agree on its sign and magnitude almost exactly (-0.422 synthetic against -0.424 real).

What is left is a *level* failure, and it is the whole of the effect. Synthetic recovery averages 0.459 against 0.772 on the real loans, so the transferred model predicts a mean of 0.457 into a book that realises 0.772—roughly 31 percentage points low—and it compresses dispersion besides (prediction SD 0.098 against a real recovery SD of 0.329). The ordering, by contrast, transfers: the transferred predictions correlate with realised real recovery at 0.400. A model that ranks correctly and is calibrated to the wrong population scores below the sample mean, which is what R^2 is measuring here. The likely cause is the population gap the paper already discloses in Section 5.6: the synthetic book is calibrated to Italian *sofferenze* recovery rates and `lgd.csv` is a general retail/mortgage LGD file recovering far more. For reference, the R^2 attainable *on* that real file is 0.1791 with `1tv` alone and 0.2903 with all three of its covariates, against the DGP-internal 0.533 that Section 3.3 shows to be a restatement of κ .

We report this as a negative result and decline to soften it: a single-covariate transfer to a differently-calibrated population is a weak test, and it failed. It converts this paper’s scaffold-grade self-labelling from a disclaimer into a measurement—but the measurement now says the protocol’s synthetic book cannot be levelled onto a foreign population, not that its `1tv` signal is absent. Track (ii) remains unexecutable here—`lgd.csv` carries no incumbent model or regulatory-capital fields—and stays pre-registered for v2.

Real-data validation pre-registration timeline.. We pre-register the v2 real-data validation with the following concrete access path and timeline.

Track A — supervisory or central-bank research-access route (preferred, 0–6 months; eligibility to be confirmed). A natural candidate is a supervisory research-access programme to a central-bank or ECB SSM NPL dataset. We have *not* verified that any specific statutory provision (for example, a CRD-related access right) grants an independent researcher access to such a database, nor that any particular central-bank academic-access policy would admit an ORCID- and SSRN-based independent-research credential; these eligibility questions remain to be confirmed before this route can be relied upon. Where such a route exists, we anticipate it would, by analogy with comparable supervisory data-access programmes, require some combination of (i) a researcher-affiliation or independent-research credential acceptable to the data holder, (ii) a registered research-protocol pre-deposit (e.g., at SSRN or arXiv), and (iii) a Data Protection Impact Assessment under GDPR Article 35; the precise requirements and review timeline are to be confirmed with the relevant authority. We expect any such dataset to need to provide a loan-level cohort of the scale required for this protocol—with key features such as district, collateral type, recovery duration, recovery amount, and securitization status—but the exact dimensions, coverage period, and access conditions are to be confirmed.

Track B — Italian NPL-servicer data-sharing agreement (6–12 months). A bilateral data-sharing agreement with a major Italian NPL servicer provides access to a production cohort on the order of $\sim 50,000$ loans with operational outcomes. Typical bilateral-agreement negotiation is 4–8 months; access typically requires pseudonymized data with on-site computation under the servicer’s secured-research environment.

Track C — Public GACS-securitization investor packs (0–3 months, smallest evidential power). The GACS-securitization market produces public investor reports with quarterly collection-rate and recovery-curve aggregates by tranche; an analyst can reverse-engineer approximate loan-level distributions but without the per-loan covariates required for the SHAP

attribution. Useful as a coarse external-validity check on the Weibull recovery-duration prediction; not sufficient for the full pipeline validation.

The v2 paper is scheduled for SSRN deposit no later than 2027-Q1 conditional on at least Track A or Track B access; the v1 protocol paper retains the synthetic-data framing in the abstract and acknowledges that the headline performance metrics are scaffold-grade pending v2.

9. Conclusion

This paper develops a comprehensive machine learning framework for Italian NPL valuation, addressing three methodological gaps: cross-validation leakage, high-cardinality categorical feature handling, and regulatory interpretability. Using purged cross-validation [31], LightGBM with native categoricals [28], and SHAP explanations [33], we demonstrate a principled approach to NPL valuation that avoids common pitfalls in financial ML.

On synthetic data the protocol behaves as designed: it recovers the injected judicial-efficiency signal, which ranks second in the SHAP attribution behind collateral `gbv_ratio`. The recovery-rate model attains a DGP-internal OOT $R^2 = 0.533$ and the AVM 0.499. We emphasise that these are DGP-recovery results, not empirical findings about real Italian NPL portfolios: the SHAP ranking reproduces calibration inputs and reorders wholesale between generator revisions (erratum item E18), a Ridge linear baseline is indistinguishable from LightGBM on this additive DGP, and the serial-dependence ablation does not establish that the purge binds (item E12).

Two claims that v1 made in this position are withdrawn rather than restated. The 4.66% Weibull round-trip against published per-district durations was a round-trip against a calibration input that no longer reproduces, and it is withdrawn as a validity check (items E9 and E10); the North–South closure gap it reported (26.6 against 65.1 months, a ratio of 2.44) is on the v2 book 13.0 and 16.7 months for a ratio of 1.28, against the 1.325 measured on 2025 court data (item E11). The gradient survives; its magnitude was roughly double what current court data support.

The one non-circular result in this paper is Section 6. Calibrating the generator on Banca d’Italia exit years 2006–2019 only leaves 20 recovery-level and 27 closure-timing points it has never seen. On those the recovery-level layer misses by 4.42 pp—the ceiling this source permits, since the best available calibration rule attains 4.43 pp—and the timing layer, refuted out of sample in v1 by 42.86 pp, now misses by 6.92 pp. That is a sixfold improvement and it is *not* a vindication: 25 of 27 points still under-predict, in the same direction as v1. The test validates the generator’s recovery *marginals* and nothing else; no loan-level result in this paper is validated by it, and none can be from a source that publishes no loan-level data. The policy and market implications above are accordingly framed as hypotheses to be tested by the pre-registered real-data release, not as conclusions of this synthetic study.

All code, data generation scripts, and model training pipelines are available in the accompanying replication package, enabling independent reproduction and extension to proprietary loan-level data.

Appendix A. Reproducibility: schema, DGP coefficients, and parameterizations

This appendix surfaces, in the text, the implementation details that are fully specified in the replication package (`synthetic_generator.py`, `models.py`, `data_acquisition.py`), which for this revision is the bundle `W_2026_08_18_P03_NPL_ML_v2`. Every file it ships is covered by its SHA256SUMS manifest, and the manifest verifies with no failures under `sha256sum -c` (text normalised to LF so the check is platform-independent); the README reproduction table maps each number to its emitting script and committed JSON. The pipeline is seeded (seed 42, LightGBM 4.6.0, SHAP 0.52.0, scikit-learn 1.9.0), which is a design property of the scripts; we do *not* claim to have re-executed the full pipeline end-to-end for this revision and therefore do not assert bit-identical regeneration as a verified fact.

A disclosure about the v1 bundle.. The v1 replication package (`W_2026_06_18_P03_NPL_ML_v1`) is retained unmodified as the historical artifact. It was never uploaded: the live record for this paper holds the manuscript PDF and no code, so no reader has ever had this package, and its manifest does not fully verify. Audited entry by entry against the repository’s object store, 72 of its 73 manifest entries reconcile under one line-ending convention or the other—41 were computed on CRLF bytes and 31 on LF bytes, so no single platform can pass the whole file—and one entry (`paper/main.bbl`) names a file the bundle does not contain. This defect predates the present revision and is disclosed rather than silently repaired, because rewriting a deposited manifest in place would misrepresent what was deposited. The v2 bundle above is sealed under a single convention and carries no such mixture.

Loan-table schema (40 columns).. The v2.1 synthetic loan table has 40 columns: 4 identifiers/dates (`loan_id`, `origination_date`, `default_date`, `split`), 2 targets (`recovery_rate`, `recovery_time_months`), and 34 modelling-eligible features. The features are `region`, `macro_zone`, `judicial_district`, `property_type`, `property_value_eur`, `superficie_sqm`, `property_condition`, `energy_class`, `collateral_value_eur`, `gross_book_value_eur`, `debtor_type`, `debtor_income_eur`, `n_creditors`, `is_secured`, `guarantee_type`, `procedure_type`, `closure_channel`, `coverage_ratio`, `vintage_years`, `forced_sale_discount`, `region_efficiency`, `nbv_gbv_ratio`, `years_since_default`, `ltv`, `pledged_share`, `accrued_interest_ratio`, and the derived `log_gbv`, `log_property_value`, `log_income`, `log_superficie`, `income_gbv_ratio`, `collateral_gbv_ratio`, `condition_score`, `energy_score`. The three columns v2 adds over v1’s 37 are `closure_channel`, `pledged_share` and `accrued_interest_ratio` (erratum item E20). “Modelling-eligible” is a schema property, not a statement about any fitted model: `closure_channel` is counted here but is deliberately *withheld* from the recovery model’s feature set, for the reason given in Section 5. The recovery-rate model uses 24 of these features and the AVM 12 (feature-count audit in `baselines.json`).

Recovery-rate DGP (11 additive components plus a per-cell offset).. The recovery target is $r_i = \text{Beta}(\alpha, \beta)$ with mean $\mu_i = \text{clip}(s_i, 0.02, 0.96)$ and concentration $\alpha + \beta = \kappa$ (so $\alpha = \kappa\mu_i$, $\beta = \kappa(1 - \mu_i)$), κ being an argument whose default is 20 (item E4; sensitivity

in Section 5.5). The signal s_i is the sum of eleven cross-section components, numbered here as they are in the source: (2) judicial efficiency $(\text{eff} - 1.5) \times 0.06$, $\text{eff} \in \{0, 1, 2, 3\}$; (3) collateralisation $(\min(\text{collateral}/\text{GBV}, 2) - 1.0) \times 0.15$ — in v2.1 $\text{collateral} = \text{property} \times \text{pledged_share}$ and $\text{GBV} = \text{property} \times \text{ltv}$, so the ratio is $\text{pledged_share}/\text{ltv}$ and moves with LTV as it should (in v1 it reduced to $1/(1 + \text{accrued})$ and measured accrued interest, item E3); (4) condition $(\text{score} - 2) \times 0.015$; (5) income capacity $\text{clip}(\log(1 + \text{income}/\text{GBV}) \times 0.03, -0.04, 0.04)$; (6) creditor count $-\min(n - 1, 6) \times 0.008$; (7) procedure type $\{\text{fallimento}: +0.03, \text{concordato}: +0.02, \text{esecuzione}: -0.01, \text{volontaria}: +0.04\}$; (8) guarantee type $\{\text{ipoteca}: +0.04, \text{pegno}: +0.02, \text{fidejussione}: +0.01, \text{none}: -0.02\}$; (9) vintage decay $-\text{years_since_default} \times 0.005$; (10) coverage $(\text{cov} - 0.55) \times 0.08$; (11) forced-sale discount $-(\text{fsd} - 0.20) \times 0.12$; (12) LTV $-(\text{ltv} - 0.68) \times \beta_{\text{ltv}}$, new in v2.1 (item E1). **The numbering starts at 2 because v1’s Component 1—an additive base of 0.50 secured against 0.18 unsecured—no longer exists.** Each collateral $\times$ channel cell is instead recentred on its Banca d’Italia Tav. A2 target, so the level sits in a per-cell offset rather than in an additive term. The injected-vs-learned mapping, including that offset row, is Table 3, which is emitted by `compute_injected_vs_learned.py` and fails loudly if the component set in the generator drifts from the one tabulated.

Recovery-timing Weibull. Recovery times are $t_i = \text{clip}(12 \cdot \text{Weibull}(k) \cdot \lambda \cdot m_i, 1, 180)$ months. The shape $k = 1.0219$ and scale $\lambda = 1.9708$ years are fitted to the Banca d’Italia Tav. 4 closure curve of entry cohort 2016 (implied median 16.5 months), *not* to per-district medians: v1 used $k = 1.5$ around a national median of 54 months derived from published district durations that do not reproduce in the current release, and that calibration was refuted out of sample by 42.86 pp (items E8 and E9). The multiplier $m_i = c_i \cdot p_i$ is normalised to book mean 1 so it cannot move the national level, where $p_i \in \{\text{esecuzione}: 1.10, \text{fallimento}: 0.90, \text{concordato}: 0.85, \text{volontaria}: 0.75\}$ and c_i is the 2025 SIECIC CEPEJ disposition-time multiplier for the loan’s district, applied to worked-out positions only ($c_i = 1$ when sold, since a sold position closes on its transfer date). Segment-level Weibull fits use `scipy.stats.weibull_min.fit` with `floc=0`.

Beta noise floor. The recovery noise concentration κ sets the irreducible R^2 ceiling via $\text{Var}(Y \mid X) = t(1 - t)/(\kappa + 1)$; this is the “sets the R^2 ceiling” row of Table 3. In v1 it was hard-coded at 20 (tightened from an earlier 12). In v2.1 it is a named argument, defaulted to 20 for comparability with the deposited number rather than because that value is defended—no public aggregate can calibrate or falsify it (Section 3.3), and Section 5.5 reports what moves with it. The generator also warns and records the affected share when κ is small enough for the Beta(0.5, 0.5) parameter floors to bind and silently override the cell calibration (item E5).

SHAP computation. SHAP values are computed with TreeExplainer on the *out-of-time test fold* (loans defaulting 2022–2025), subsampled to at most 5,000 observations with a fixed `RandomState(42)` (`models.py`, `_train_lgb_model`, `compute_shap=True`). The reported mean-|SHAP| ranks in Table 9 are therefore OOT-fold attributions, not training-fold.

Share-of-wallet leakage exclusion. The share-of-wallet target income/GBV is a deterministic function of `debtor_income_eur` and `gross_book_value_eur`; these and their logs are

excluded from the feature set to avoid a tautological $R^2 \approx 0.98$, and the model is fit on exogenous covariates only (`models.py`, `train_share_of_wallet`).

External benchmark data availability (added 2026-08-10).. The real-data LGD benchmark used in Section 5.6 is `lgd.csv` ($n = 2,545$ defaulted loans; 191,134 bytes; SHA-256 `2e3f964a88bc97f6...`), the companion dataset of Baesens et al. [6], available from the book’s companion website at <http://www.creditriskanalytics.net>. Its redistribution licence is unstated, so the file is *not* shipped inside this replication package. Instead, `code/data/external/fetch_lgd.py` downloads it from the pinned source URL (or verifies a locally supplied copy) and checks the full SHA-256 recorded in the script before any use; `code/data/external/README.md` records the complete provenance note.

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